# Process Analysis

## Lecture 06

BUSINESS PROCESS MANAGEMENT

**IS 2206**

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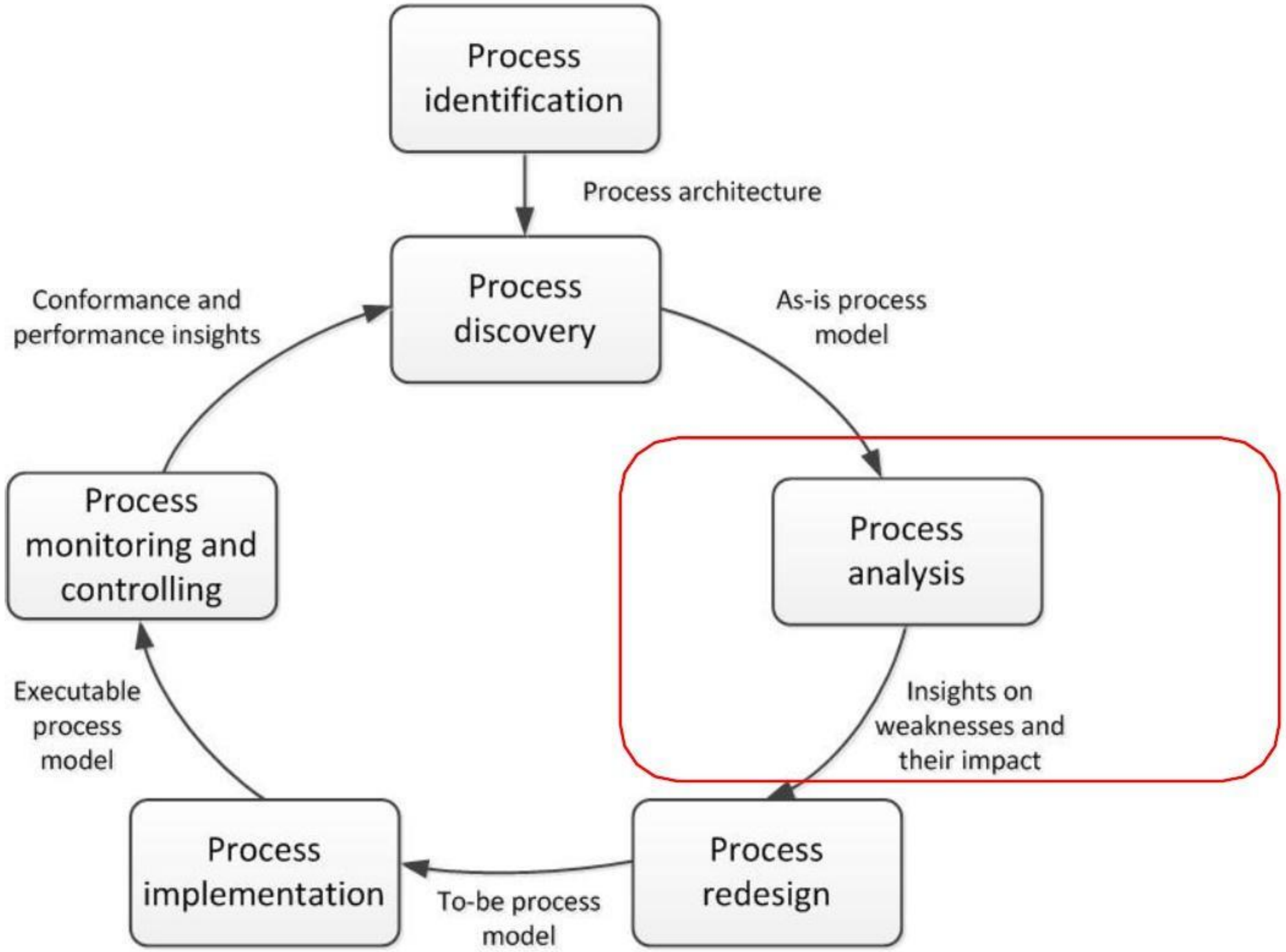

Process identification
Process architecture
Process discovery
As-is process model
Process analysis
Insights on weaknesses and their impact
Process redesign
To-be process model
Process implementation
Executable process model
Process monitoring and controlling
Conformance and performance insights

# OUTLINE

1. What is Process Analysis?
2. Qualitative Process Analysis
3. Quantitative Process Analysis

# TOPIC 01

What is Process Analysis?

# 1.1 What is Process Analysis

A business may consist of several business process And the performance of a business is highly influenced by these Business Process. Thus it is essential to do a process analysis in order to assess whether there is a need for Business Process Re-Engineering.

**The objective of the process analysis is to figure out :**

**" Whether the current business processes aligned with the goals and objectives of the organization"**

- Business process analysis (BPA) is a methodology to understand the health of different operations within a business to improve process efficiency.
- Without a proper analysis, your teams will waste a lot of time and effort in solving the wrong problems.
- Many techniques are available to understand what is happening in the current (As-Is) process model. This analysis aims to identify weaknesses in the current process and the impact this may be having on the business outcomes.

# 1.2 Process Analysis Techniques

**Qualitative Analysis**

- Value-Added Analysis
- Root-Cause Analysis
- Pareto Analysis
- Issue Register

**Quantitative Analysis**

- Quantitative Flow Analysis
- Queuing Theory
- Process Simulation

# 1.3 Qualitative Vs. Quantitative Analysis

- Qualitative process analysis provides systematic insights into a process.
- But the results are not detailed enough to provide a solid basis for decision making
- To make estimates in terms of costs, time (etc.) we need to go beyond Qualitative Analysis.
- Thus Quantitative Analysis techniques identify ways of analyzing a business process quantitatively in terms of performance measures such as cycle time, total waiting time and cost.

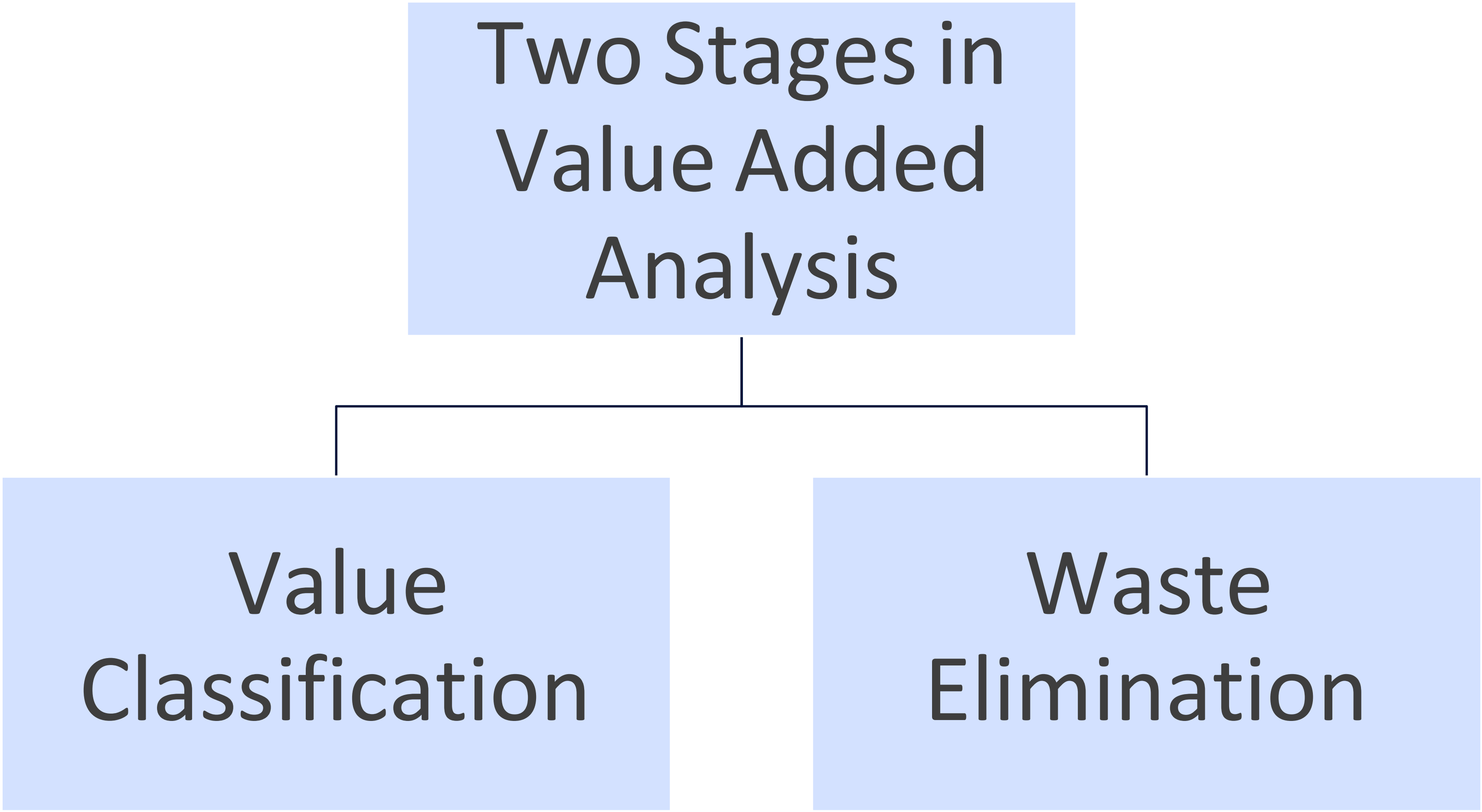
# 1.4 Purpose of Qualitative Process Analysis

- To identify and understand all the issues involved, and prioritize them accordingly.
- To identify wastes, redundancies, or losses incurred in the processes and eliminate them.

# TOPIC 02

Qualitative Process Analysis

# 2.1 Value-Added Analysis



# 2.1.1 Value Added Analysis - Value Classification

1. Decompose the process in to steps
   - Steps performed before a task
   - The task itself, can be decomposed into smaller steps
   - Steps performed after a task But the results are not detailed enough to provide a solid basis for decision making
2. Classify each step as
   - *Value-adding (VA):* Produces value or satisfaction to the customer.
   - *Business value-adding (BVA):* Necessary or useful for the business to run smoothly, or required due to the regulatory environment, e.g. checks, controls.
   - *Non-value-adding (NVA):* everything else including handovers, delays and rework

# 2.1.1 Value Added Analysis - Value Classification (Contd..)

## *Value Adding (VA)*

Produces value or satisfaction to the customer.

Criteria :

If the step is removed, would the customer perceive that the end product or service is less valuable?

Is the customer willing to pay for this step?

Would the customer agree that this step is necessary to achieve their goals?

Examples:

- Order-to-cash process: Confirm delivery date, Deliver products
- University admission process: Assess application, Notify admission outcome

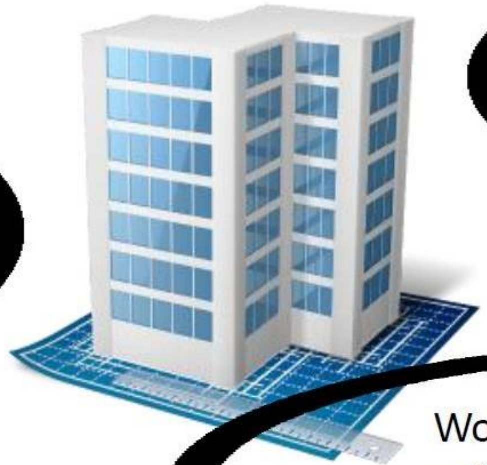
# 2.1.1 Value Added Analysis - Value Classification (Contd..)

## *Business Value Adding (BVA)*

Necessary or useful for the business to run smoothly, or required due to the regulatory environment ( E.g. checks, controls)

Criteria

Is this step required in order to comply with regulatory requirements?



Is this step required in order to collect revenue, to improve or grow the business?

Would the business (potentially) suffer in the long-term if this step was removed? Does it reduce risk of business losses?

Example :

- Order-to-cash process: Check purchase order, Check customer's credit worthiness, Issue invoice, Collect payment, Collect customer feedback
- University admission process: Verify completeness of application, Check validity of degrees, Check validity of language test results

# 2.1.1 Value Added Analysis - Value Classification (Contd..)

## *Non Value Adding (NVA)*

Everything else  other than VA and BVA

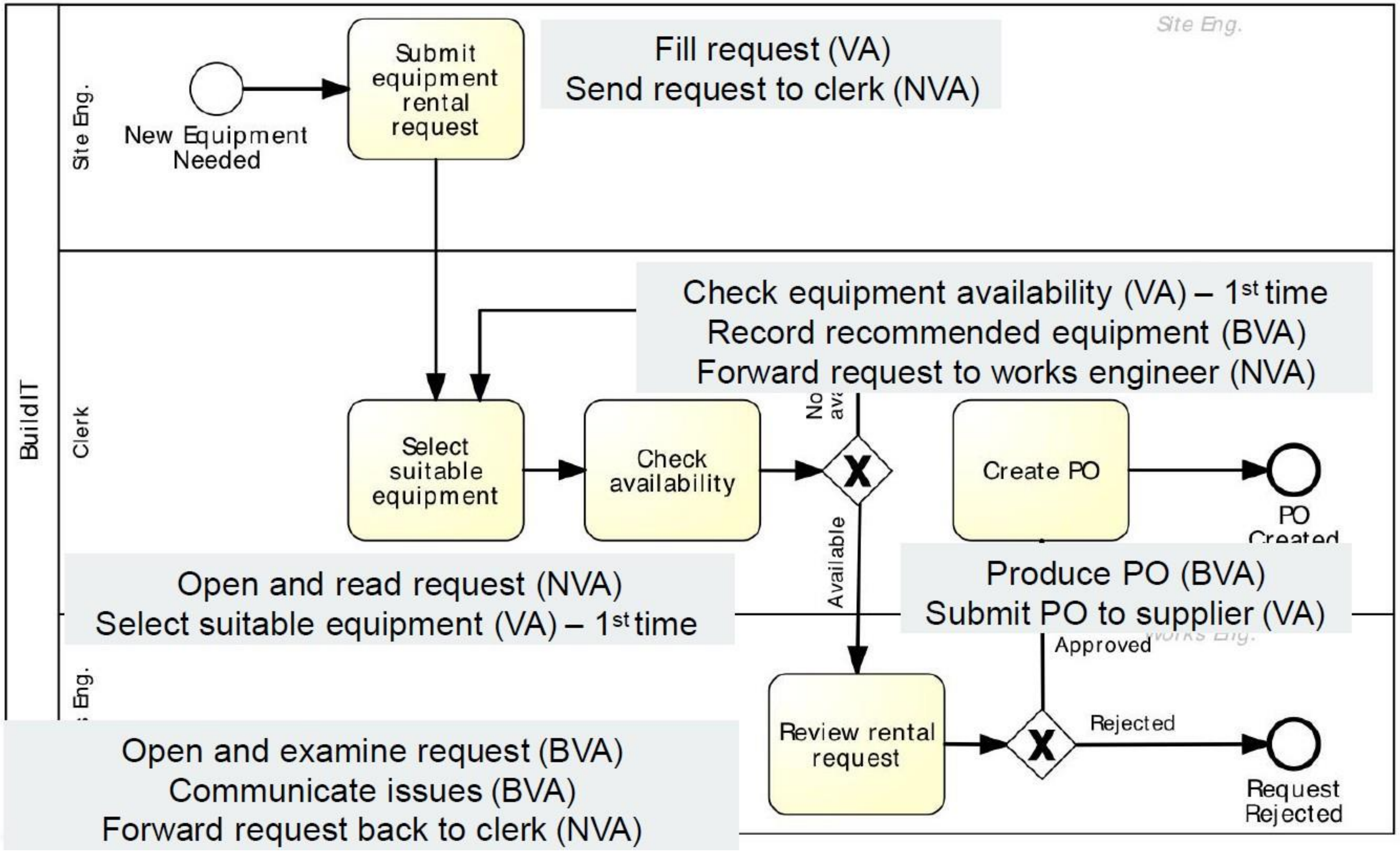
Including handovers, delays and rework, waiting times , defect correction

Examples

- Order-to-cash: Forward PO to warehouse, Re-send confirmation,  Receive rejected products
- University admission: Forward applications to committee, Receive  admission results from committee

# 2.1.1 Value Added Analysis - Value Classification (Contd..)

- *Example:* Extract of Equipment Rental Process



# 2.1.1 Value Added Analysis - Value Classification (Contd..)

- *Example:* Equipment Rental Process –VAAnalysis

| Step | Performer | Classification |
|---|---|---|
| Fill request | Site engineer | VA |
| Send request to clerk | Site engineer | NVA |
| Open and read request | Clerk | NVA |
| Select suitable equipment | Clerk | VA |
| Check equipment availability | Clerk | VA |
| Record recommended equipment & supplier | Clerk | BVA |
| Forward request to works engineer | Clerk | NVA |
| Open and examine request | Works engineer | BVA |
| Communicate issues | Works engineer | BVA |
| Forward request back to clerk | Works engineer | NVA |
| Produce PO | Clerk | BVA |
| Send PO to supplier | Clerk | VA |

# 2.1.2 Value Added Analysis – Waste Elimination

*"All we are doing is looking at the time line, from the moment the customer gives us an order to the point when we collect the cash. And we are reducing the time line by reducing the* ***non- value-adding*** *wastes"*

*-Taiichi Ohno*

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

1. Unnecessary Transportation (send, receive)
2. Motion (drop-off, pick-up, go to)

**Move**

3. Inventory (large work-in-process)
4. Waiting (waiting time between tasks)

**Hold**

5. Over-Processing (performing what is not yet needed or might not be needed)
6. Over-Production (unnecessary cases)
7. Defects (rework to fix defects)

**Over do**

8. Resource underutilization (idle resources)

Source: Seven Wastesdefinedby Taiichi Ohno 8th waste coinedby Ben Chavis, Jr.

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *General Practices*

- Minimize or eliminate NVA steps
- Some NVA steps can be eliminated by automation. Automation can bring transparency into the process to its performers
- Eliminate NVA performers (e.g. Remove Clerk in the Equipment Rental Process and give work to the Site Engineer, reduces handovers)
- Eliminate the need of NVA steps

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

### 1. Unnecessary Transportation

- Send or receive materials or documents (incl. electronic) taken as input or output by the process

Example:

To apply for admission at a University, students fill in an online form. When a student submits the online form, a PDF document is generated. The student is requested to download it, sign it, and send it by post together with the required documents: 1. Certified copies of degree and academic transcripts. 2. Results of language test. 3. CV.
When the documents arrive to the admissions office, an officer checks their completeness. If a document is missing, an e-mail is sent to the student. The student has to send the missing documents by e-mail or post depending on document type.

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

### 2. Motion

- Motion of resources internally within the process
- Common in manufacturing processes, less common in business processes

Examples

- Vehicle inspection process, a process worker moves with the inspection forms from one inspection base to another; in some cases inspection equipment also needs to be moved around
- Approval process, a process workers moves around the organization to collect signatures

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

### 3. Inventory

- Materials inventory
- Work--in--process (WIP)

Examples

- Vehicle inspection process, when a vehicle does not pass the first inspection, it is sent back for adjustments and left in a pending status. At a given point in time, about 100 vehicles are in the "pending" status across all inspection stations

- University admission process: About 3000 applications are handled concurrently

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

### 4. Waiting

- Waiting for materials or input data
- Task waiting for a resource
- Resource waiting for work (resource idleness)

Examples

- Vehicle inspection process: A technician at a base of the inspection station waiting for the next vehicle
- Approval process: Request waiting for approver
- University admission process: Incomplete application waiting for additional documents; batch of applications waiting for committee to meet

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

### 5. Defects

- Correcting or compensating for a defect
- Rework loops

Examples

- Vehicle inspection: A vehicle needs to come back to a station due to an omission
- Travel approval: Request sent back to requestor for revision
- University admission: Application sent back to applicant for modification; request needs to be re--assessed later due to incomplete information

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

### 6. Over-Processing

- Tasks performed unnecessarily given the outcome of the process
- Unnecessary perfectionism

Examples

- Vehicle inspection: Technicians take time to measure vehicle emissions with higher accuracy than required, only to find that the vehicle clearly does not fulfill the required emission levels
- Travel approval: 10% of approvals are trivially rejected at the end of the process due to lack of budget
- University admission: Officers spend time verifying the authenticity of degrees, transcripts and language test results. In 1% of cases, these verifications uncover issues.
  Verified applications are sent to the admissions committee. The admission committee accepts 20% of the applications it receives

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

### 7. Over-Production

- Unnecessary process instances are performed, producing outcomes that do not add value upon completion

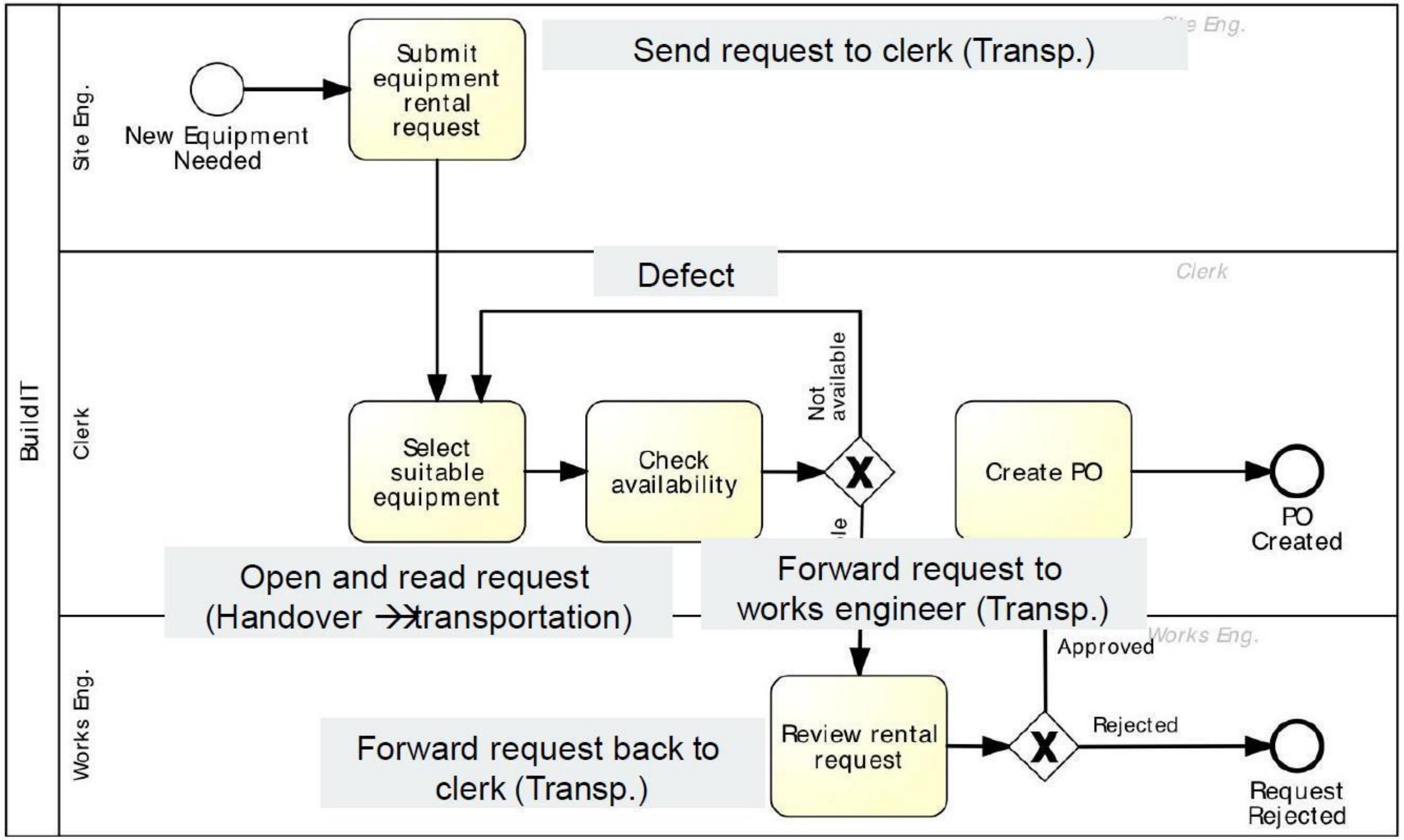
Examples

- Order--to--cash: In 50% of cases, issued quotes do not lead to an order
- Travel approval: In 5% of cases, travel requests are approved but the travel is cancelled
- University admission: About 3000 applications are submitted, but only 800 are considered eligible after assessment

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

- *Example:* Equipment Rental Process - Waste



# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

- *Example:* Equipment Rental Process – Waste (Contd..)

**Transportation**

- Site engineer sends request to clerk
- Clerk forwards to works engineer
- Works engineer send back to clerk

**Inventory**

- Equipment kept longer than needed

**Waiting**

- Waiting for availability of works engineer to approve

# 2.1.2 Value Added Analysis – Waste Elimination(Contd..)

## *7+1 Sources of Waste*

- *Example:* Equipment Rental Process – Waste (Contd..)

**Defect**

- Selected equipment not available, alternative equipment sought
- Incorrect equipment delivered and returned to supplier

**Over--processing**

- Clerk finds available equipment and rental request is rejected because equipment not needed
- Rental requests being approved and then canceled by site engineer

**Over--production**

- Equipment being rented and not used at all

# 2.2 Root-Cause Analysis

- A family of techniques to help analysts to identify and understand root cause(s) of problems or undesirable events.
- Commonly used in context of an accident or incident analysis
- Also used in manufacturing processes to understand the root cause of defects in a product.
- In the context of business process analysis root cause analysis is helpful to understand and identify issues that prevent a process from having a better performance.
- Root cause analysis techniques
    - Cause-Effect Diagrams
    - Why-why Diagrams

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams (Fishbone/Ishikawa)

- Helps to identify all the likely causes of a problem. Depicts the relationship between a given negative effect and its causes.
- Negative effect is
  - A recurrent issue
  - Undesirable level of process performance
- Causes can be divided into causal and contributing factors
- In a cause-effect diagram factors are grouped into categories and possibly sub categories

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Casual Vs. Contributing Factors*

**CAUSAL FACTORS**

Factors that, if corrected, eliminated or avoided would prevent the issue from occurring.

E.g. In the context of insurance claims handling process **errors in the estimation of damages** lead to incorrect claim assessment. If the damage estimation issues were eliminated, a number of occurrences of the issue "Incorrect claim assessment" would be prevented.

**CONTRIBUTING FACTORS**

Factors that set the stage for, or that increase the chances of a given issue occurring.

E.g. In the user interface for lodging the insurance claims requires a claimant to enter a few date (e.g. date when the incident occurred), but the interface does not provide a calendar widget so that the user can easily select the date. This deficiency in the user interface may increase the chances of erroneous entering of the date by the user. Thus, it contributes to the issue "Incorrect data entry"

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Steps of Drawing a Cause-Effect Diagram*

- Main Steps
  1. **Step 1:** Identify the Problem
  2. **Step 2:** Work Out the Major Factors Involved
  3. **Step 3:** Identify Possible Causes
  4. **Step 4:** Analyze Your Diagram

Source: MindTools(http://www.mindtools.com/pages/article/newTMC_03.htm)

- Categorization for cause-effect analysis (6M's)
  - ✓ **Machine**
  - ✓ **Method**
  - ✓ **Material**
  - ✓ **Man**
  - ✓ **Measurement**
  - ✓ **Milieu**
- 6M's are usually used for the manufacturing industry
- This categorization act as guidelines for brainstorming during root cause analysis

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Steps of Drawing a Cause-Effect Diagram*

1. **Machine**
   - Factors relating to the technology used
     - E.g. software failures, network failures, system crashes occurs in the information system that supports a business process
   - Possible Sub-categories:
     - Lack of functionality in application system
     - Redundant storage of data across systems, leading double data entry
     - Low performance of IT of network systems, leading for (e.g) low response time
     - Poor user interface designs, leading for erroneous data entry and missing data (Some necessary fields are Missing !!)
     - Lack of integration between multiple systems within the enterprise or with external systems

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Steps of Drawing a Cause-Effect Diagram*

2. **Method**
    - Factors stemming from the way a process is defined, understood or performed
        - E.g. In a given process when participant A thinks participant B will send an email to the customer but participant B does not send is because he is not aware that he has to send it
    - Possible Sub-Categories
        - Unclear, unsuitable or inconstant assignment of decision making and processing responsibilities to process participants
        - Lack of empowerment of process participants
        - Lack of timely communication between process participants or process participants and the customer

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Steps of Drawing a Cause-Effect Diagram*

3. **Material**
   - Factors stemming from the raw materials, consumables or data required as input by the activities in the process
     - E.g. Incorrect data leading to wrong decision, being made during the execution of a process

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Steps of Drawing a Cause-Effect Diagram*

4. **Man**
   - Factors related to wrong assessment or incorrectly performed step
     - E.g. claims handler incorrectly accepting a claim even though data in the claim and the rules used for assessing the claim require that the claim be rejected.
   - Possible sub-categories
     - Lack of training and clear instructions for process participants
     - Lack of incentive system to motivate process participants sufficiently
     - Expecting too much from the process participants (e.g. overly hectic schedules)
     - Inadequate recruitment of process participants.

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Steps of Drawing a Cause-Effect Diagram*

5. **Measurement**
   - Factors related to measurements or calculations made during the process.
     - E.g. In context of an insurance claim, where the amount to be paid to the customer is miscalculated due to an inaccurate estimation of the damage being claimed.

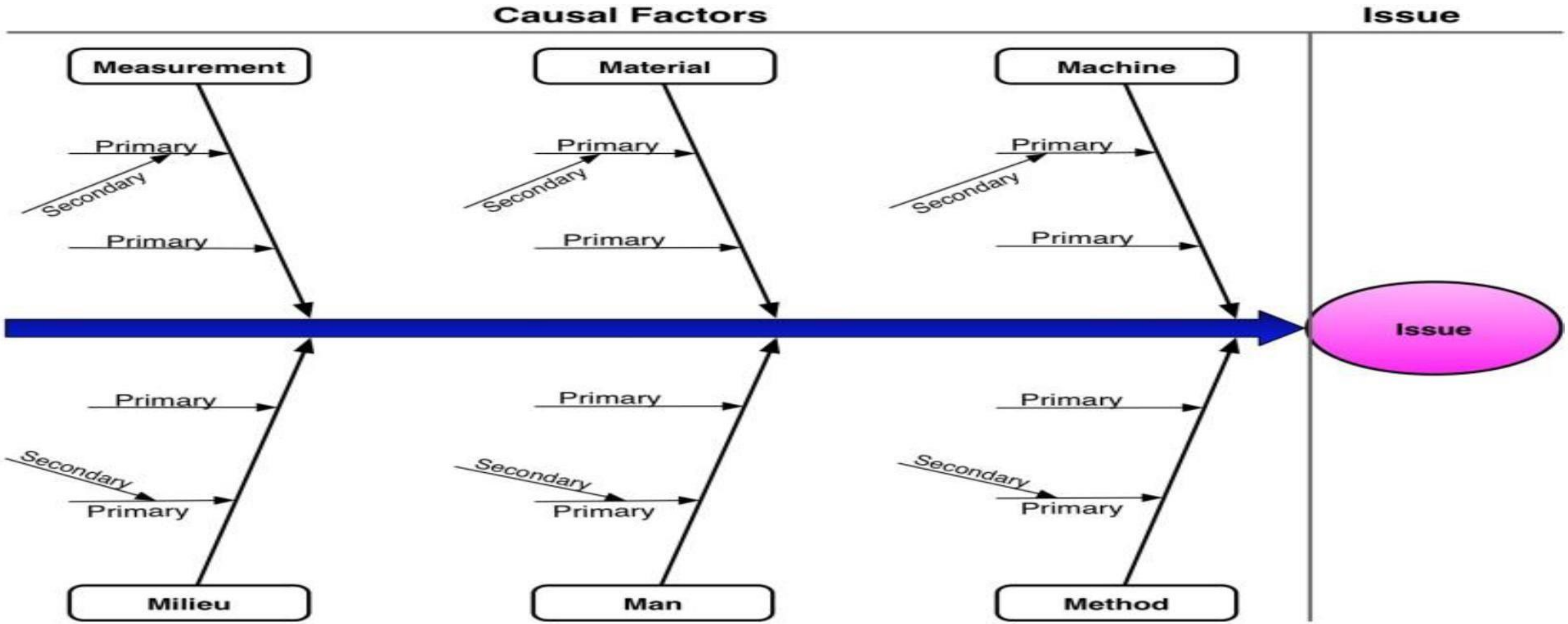
# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Steps of Drawing a Cause-Effect Diagram*

6. **Milieu (Setting)**
   - Factors stemming from environment in which the process is executed
   - Milieu conditions are outside the control of the process participants, process owner and other company managers
     - E.g. factors originating from the customer, supplier or other external actors
   - Possible sub-categories
     - originating actor is a possible sub categorization

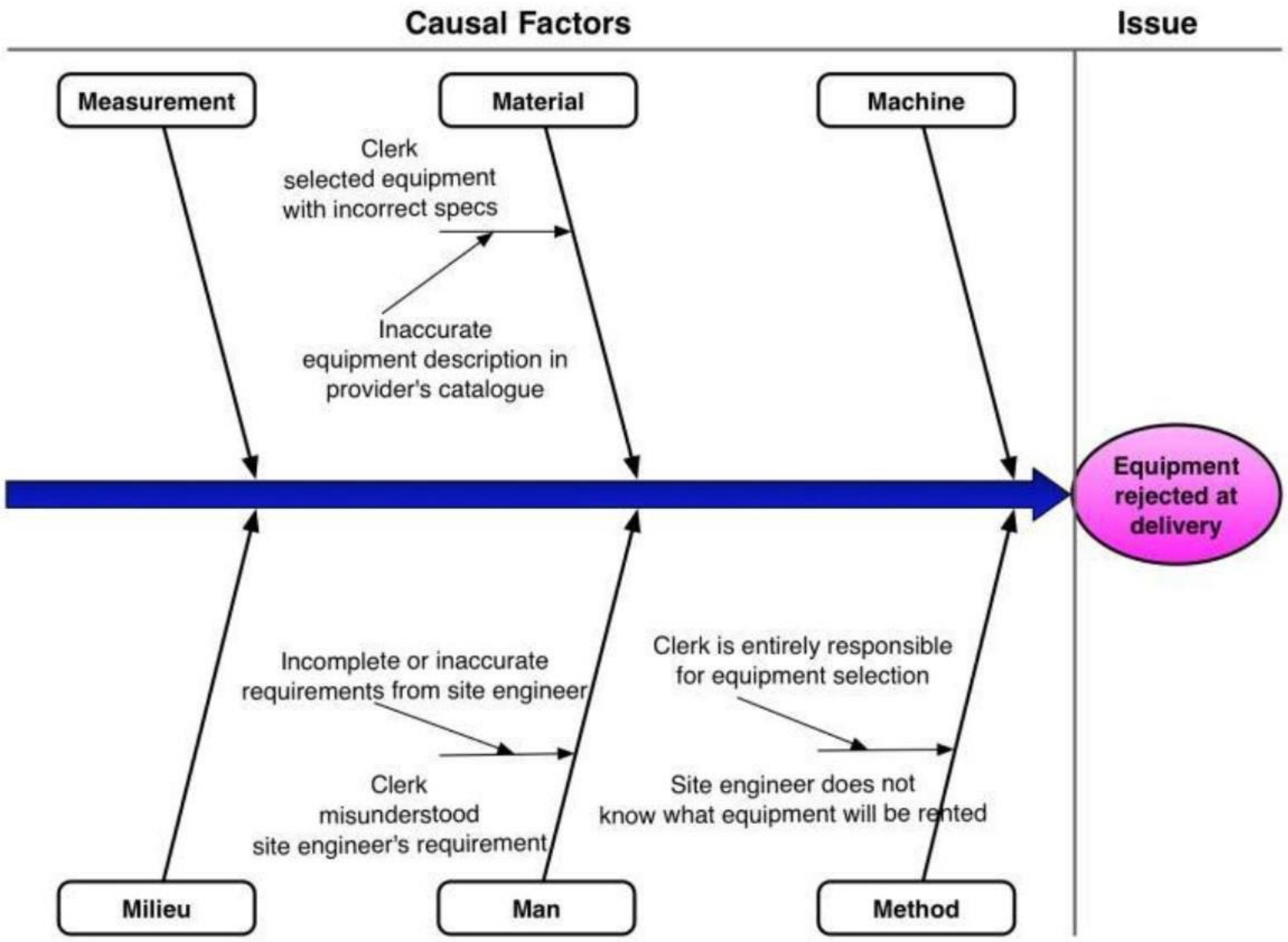
# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Structure*



# 2.2.1 Root Cause Analysis – Cause Effect Diagrams
## *Examples*

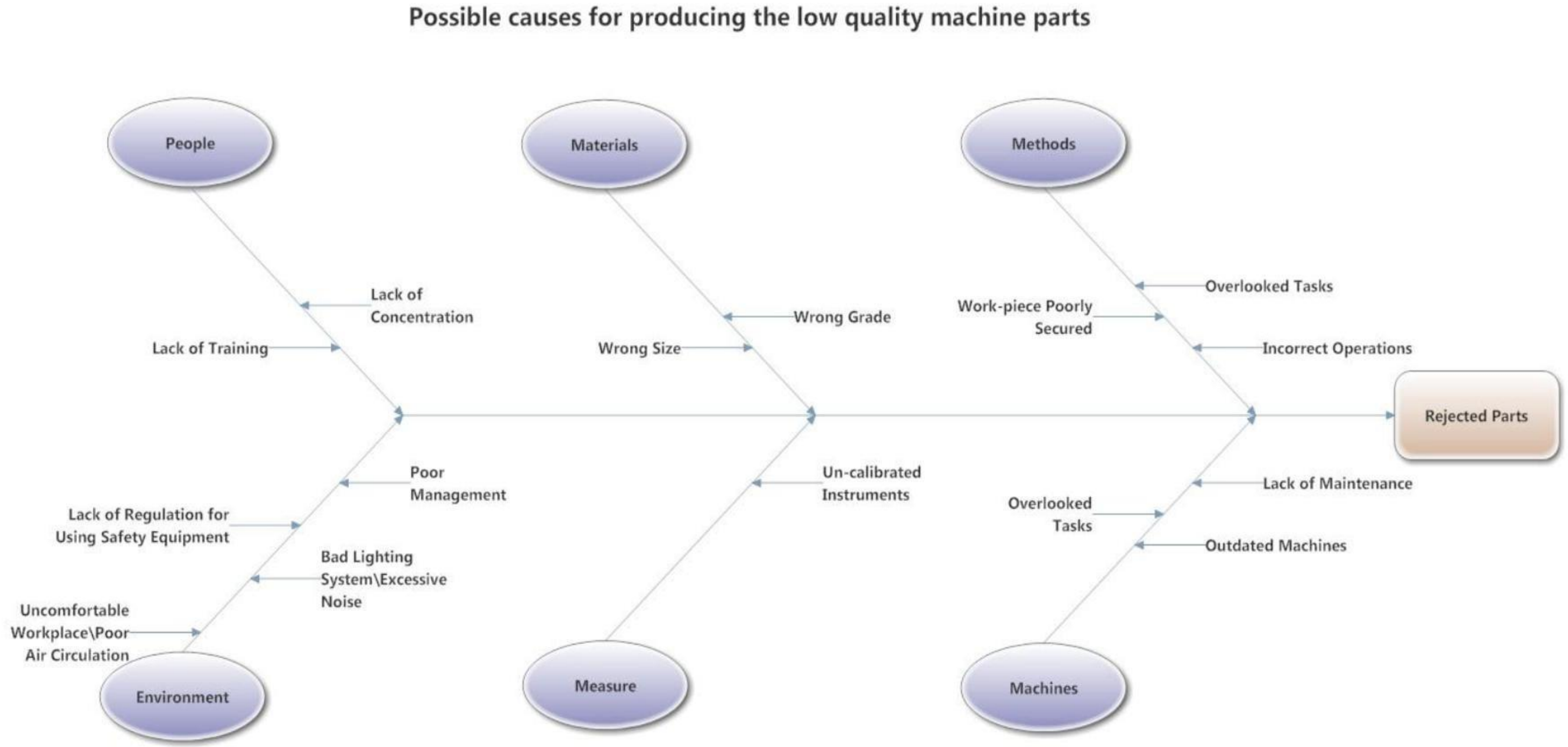
- **Rejected Equipment**



# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Examples*

- **Low Quality Machine Parts**



# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Exercise*

### ❖ Admission Process - Scenario

Consider the following process for the admission of graduate students at a university. In order to apply for admission, students first fill in an online form. Online applications are recorded in an information system to which all staff members involved in the admissions process have access to. After a student has submitted the online form, a PDF document is generated and the student is requested to download it, sign it, and send it by post together with the required documents, which include: 1. Certified copies of previous degree and academic transcripts. 2. Results of English language test. 3. Curriculum vitae.

- When these documents are received by the admissions office, an officer checks the completeness of the documents. If any document is missing, an e-mail is sent to the student. The student has to send the missing documents by post. Assuming the application is complete, the admissions office sends the certified copies of the degrees to an academic recognition agency, which checks the degrees and gives an assessment of their validity and equivalence in terms local education standards. This agency requires that all documents be sent to it by post, and all documents must be certified copies of the originals. The agency sends back its assessment to the university by post as well. Assuming the degree verification is successful, the English language test results are then checked online by an officer at the admissions office. If the validity of the English language test results cannot be verified, the application is rejected (such notifications of rejection are sent by e-mail). Once all documents of a given student have been validated, the admission office forwards these documents by internal mail to the corresponding academic committee responsible for deciding whether to offer admission or not. The committee makes its decision based on the academic transcripts and the CV. The committee meets once every 2 to 3 weeks and examines all applications that are ready for academic assessment at the time of the meeting.
- At the end of the committee meeting, the chair of the committee notifies the admissions office of the selection outcomes. This notification includes a list of admitted and rejected candidates. A few days later, the admission office notifies the outcome to each candidate via e-mail. Additionally, successful candidates are sent a confirmation letter by post.

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Exercise*

❖ **Admission Process – Scenario Continued…**

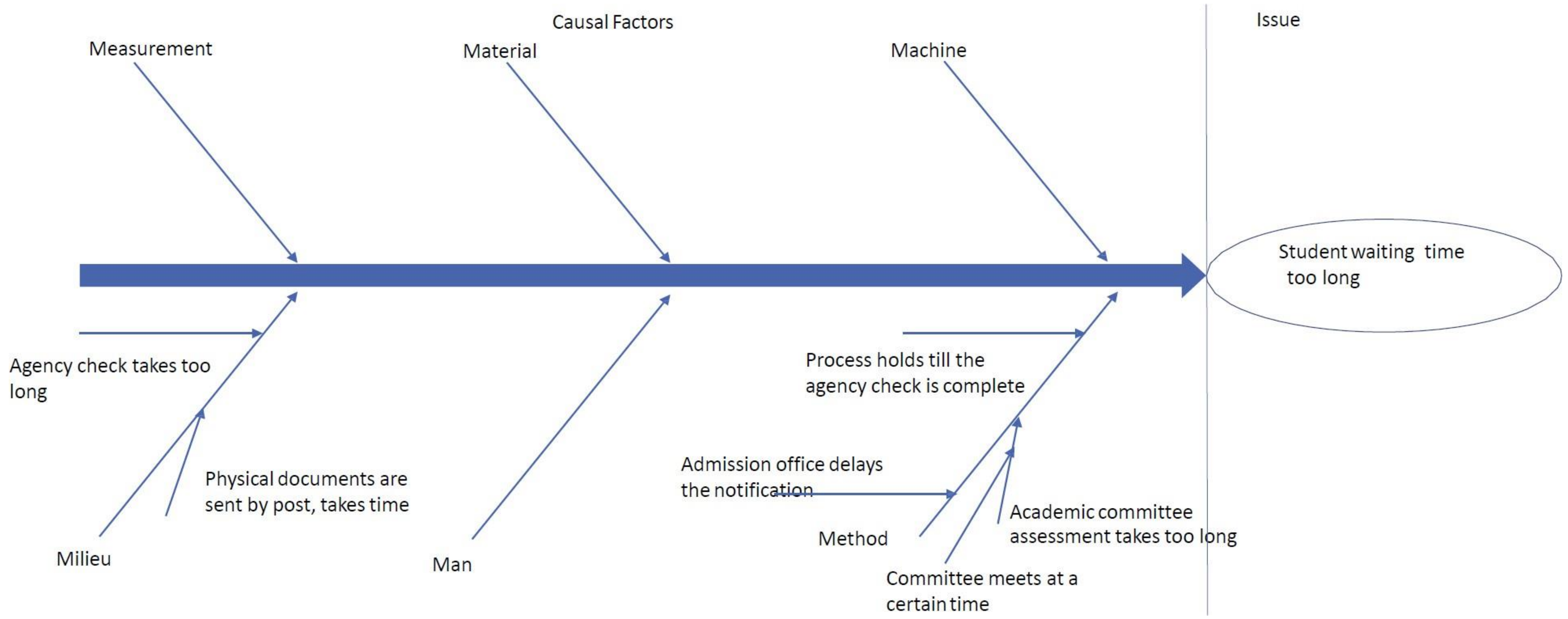
- Consider the above process. One of the issues faced by the university in the process is that students have to wait too long to know the outcome of the application (especially for successful outcomes). It often happens that by the time a student is admitted the student had decided to go to another university (students normally sends multiple applications to multiple universities).

- Analyze the causes of this issue using a cause-effect diagram.

# 2.2.1 Root Cause Analysis – Cause Effect Diagrams

## *Exercise*

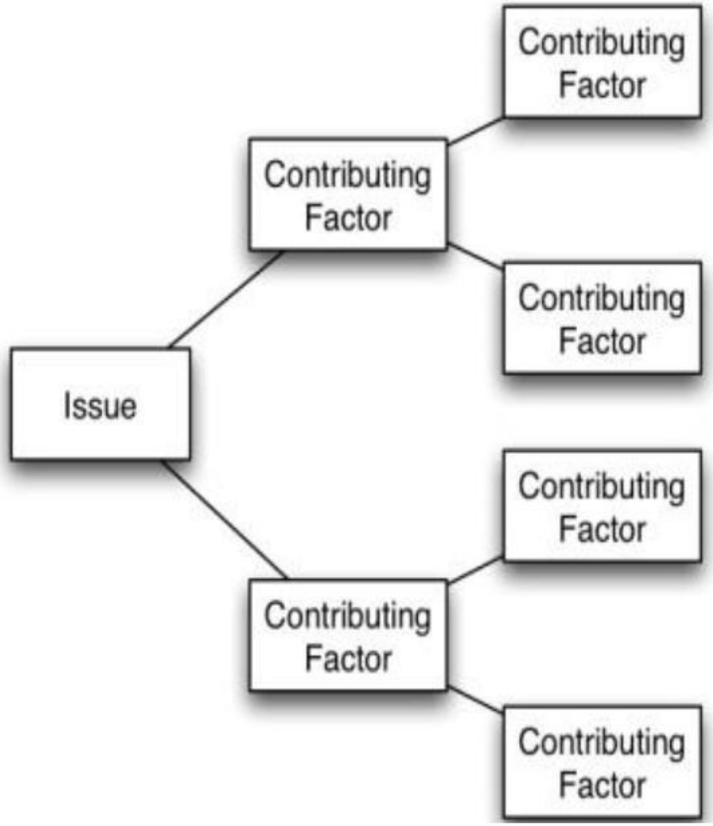
❖ **Admission Process - Answer**



# 2.2.2 Root Cause Analysis – Why Why Diagrams

## *Introduction*

- Also known as tree diagrams
- Technique to analyze causes of negative effects
- Developing the tree diagram helps you move your thinking step by step from generalities to specifics.
- The basic idea is to reclusively ask the question –“Why has something happened?”
- This question is asked multiple times until a factor that stakeholders perceive to be a root cause is found.



# 2.2.2 Root Cause Analysis – Why Why Diagrams

## *When to use a Why-Why diagram*

- When an issue is known or being addressed.
- When developing actions to carry out a solution or other plan.
- When analyzing processes in detail.
- When probing for the root cause of a problem.
- When evaluating implementation issues for several potential solutions.
- As a communication tool, to explain details to others.

# 2.2.2 Root Cause Analysis – Why Why Diagrams

## *Example*

**Equipment Rental**

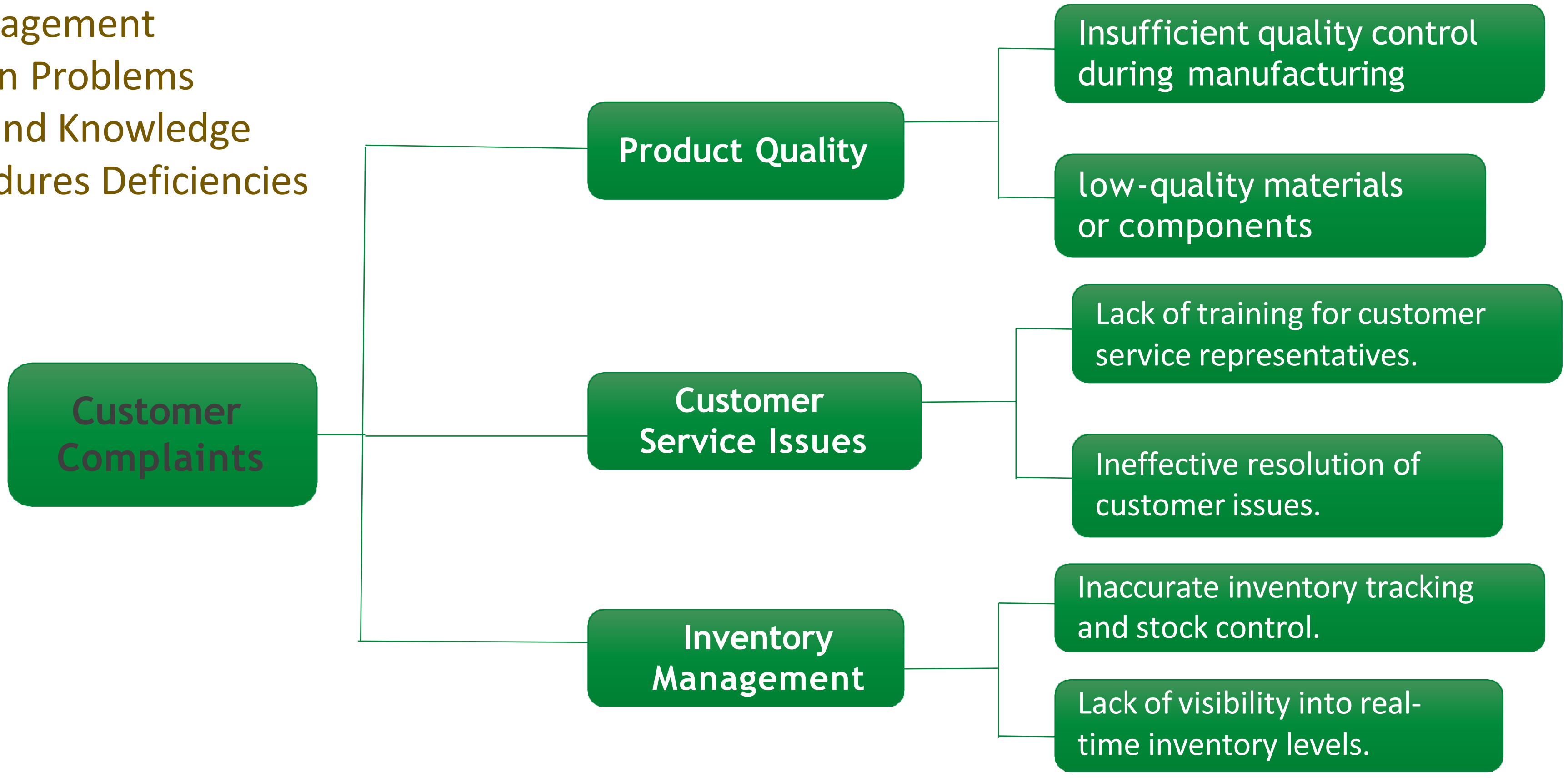
- Site engineers keep equipment longer, why?
- Site engineer fears that equipment will not be available later when needed, why?
- Time between request and delivery too long, why?
- Excessive time spent in finding a suitable equipment and approving the request, why?
- Time spent by clerk contacting possibly multiple suppliers sequentially;
- Time spent waiting for works engineer to check the requests;

# 2.2.2 Root Cause Analysis – Why Why Diagrams

## *Example*

- Increasing Customer Complaints at a Retail Store

**Major Factors :** Product Quality
Customer Service Issues
Inventory Management
Communication Problems
Staff Training and Knowledge
Policy & Procedures Deficiencies



# 2.2.2 Root Cause Analysis – Why Why Diagrams

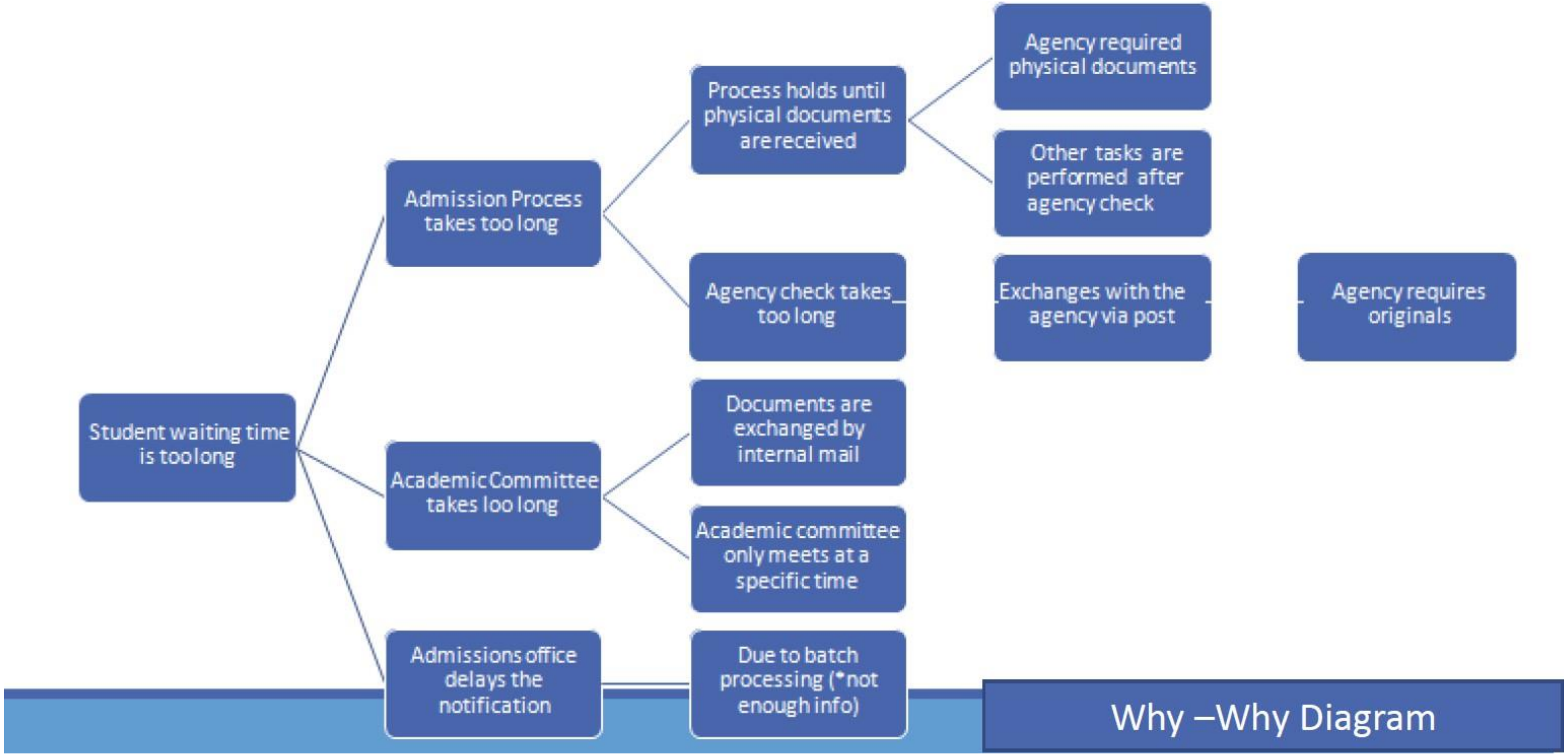
## *Exercise*

❖ **Admission Process**

- Consider the Admission process.
- Draw a Why-Why Diagram for the issue of Student waiting time is too long .

# 2.2.2 Root Cause Analysis – Why Why Diagrams *Exercise*

- **Admission Process - Answer**



# Issue Documentation & Impact Assessment

- **Once a root cause analysis identifies the factors behind a given issue building an understanding of these issues is critical to prioritize the issues.**
- Attention can the be put more onto the issues based on the prioritization
- Complementary techniques for impact assessment
  - Issue Register
  - Pareto Analysis and PICK Charts

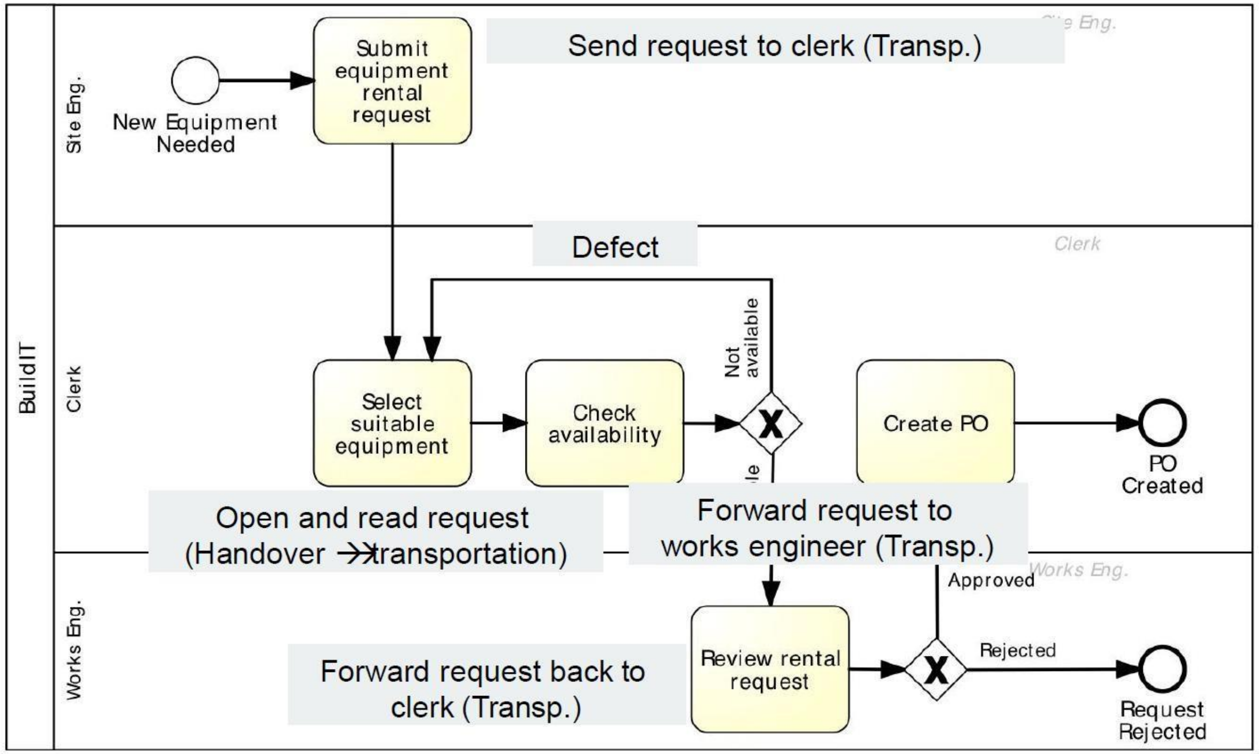
# 2.3 Issue Register

- After identifying problems in a business process, we need to record them in an **Issue Register**.
- Purpose: To categorize identified issues as part of as-is process modelling
- Usually a table with the following columns (possibly others):
  - ✓ Issue number
  - ✓ Name
  - ✓ Description/explanation
  - ✓ Impact: Qualitative vs. Quantitative
  - ✓ Possible solution

# 2.3 Issue Register
## *Example*

- **Equipment Rental Process**

Equipment rental process: wastes



# 2.3 Issue Register - *Example*

## ❖ Equipment Rental Process – Issue Register

| Name | Explanation | Assumptions | Qualitative Impact | Quantitative Impact |
|---|---|---|---|---|
| Equipment kept longer than needed | Site engineers keep the equipment longer than needed by means of deadline extensions | BuildIT rents 3000 pieces of equipment p.a. In 10% of cases, site engineers keep the equipment two days longer than needed. On average, rented equipment costs 100 per day | | $0.1 \times 3000 \times 2 \times 100 =$ 60,000 p.a. |
| Rejected equipment | Site engineers reject delivered equipment due to non-conformance to their specifications | BuildIT rents 3000 pieces of equipment p.a. Each time an equipment is rejected due to an internal mistake, BuildIT is billed the cost of one day of rental, that is 100. 5% of them are rejected due to an internal mistake | Disruption to schedules. Employee stress and frustration | $3000 \times 0.05 \times 100 =$ 15,000 p.a. |
| Late payment fees | BuildIT pays late payment fees because invoices are not paid by the due date | BuildIT rents 3000 pieces of equipment p.a. Each equipment is rented on average for 4 days at a rate of 100 per day. Each rental leads to one invoice. About 10% of invoices are paid late. Penalty for late payment is 2%. | | $0.1 \times 3000 \times 4 \times 100 \times 0.02 = 2400$ p.a. |

- This allows the organization to compare different issues and decide which improvements should be given the highest priority.

# 2.4 Pareto Analysis

- Useful to prioritize a collection of issues or factors behind an issue
- The Pareto principle, also known as the 80/20 rule, suggests that roughly 80% of effects come from 20% of causes.
- In BPM, this means that a small number of issues or process steps might be responsible for the majority of problems or inefficiencies.
- By focusing on the "vital few" causes that contribute to the majority of problems, organizations can streamline processes, reduce costs, and improve overall efficiency.
- Steps:
  - Identify the problem or problems.
  - List or identify the cause of the issues or problems, noting that there could be multiple causes.
  - Score the problems by assigning a number to each one that prioritizes the problem based on the level of negative impact on the company.
  - Organize the problems into groups. Eg- customer service or system issues.
  - Develop and implement an action plan, focusing on the higher-scored problems first, and solve the problems.

# 2.4 Pareto Analysis

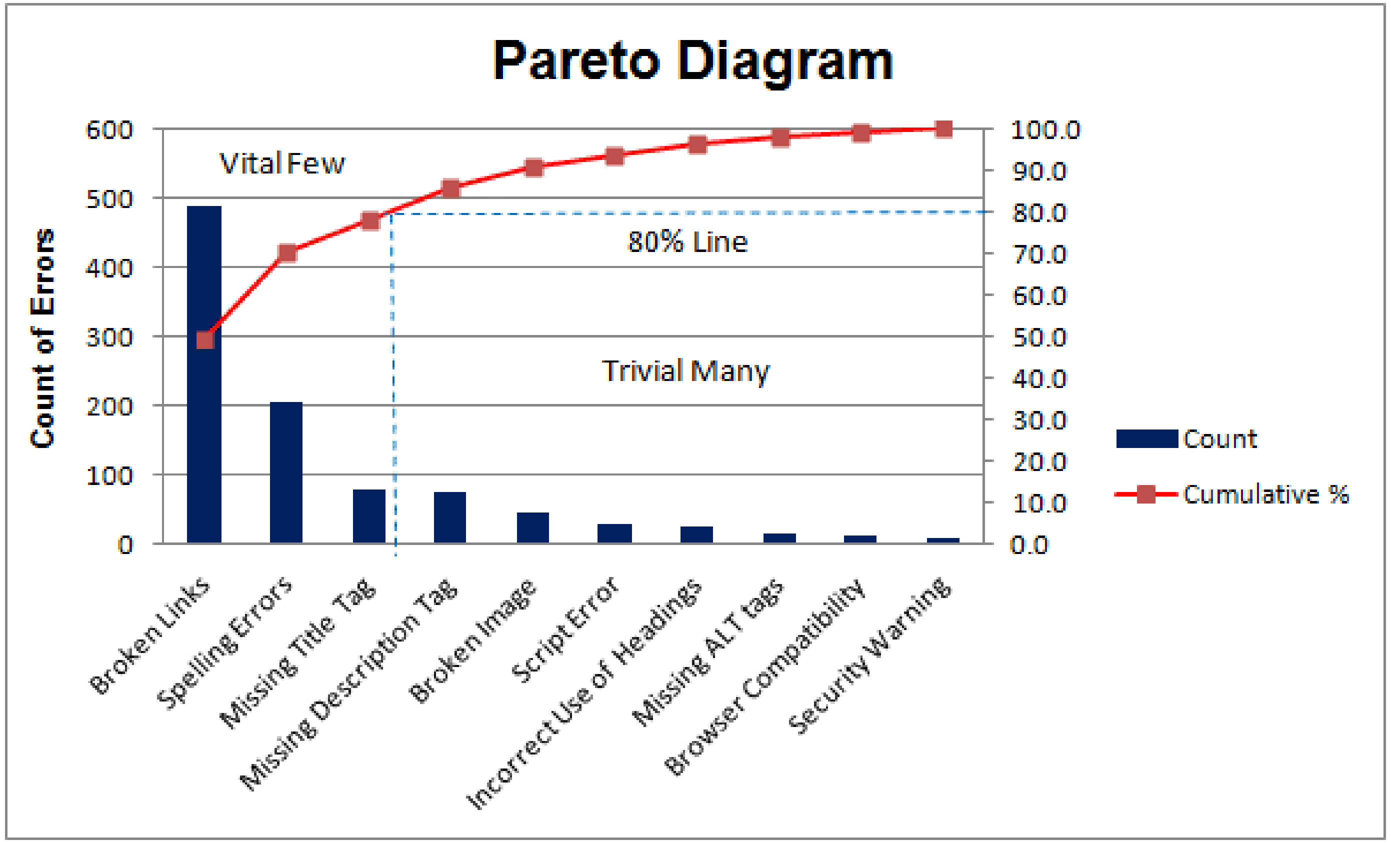
- Process of making a Pareto chart

1. Develop a list of problems to be compared.
2. Develop a standard measure for comparing the items—for example, how often it occurs: frequency (e.g., utilization, complications, errors); how long it takes (time); and how many resources it uses (cost).
3. Choose a time frame for collecting the data.
4. For each item, tally how often it occurred (or cost or total time). Then, add these amounts to determine the total for all items.
5. Find the percent of each item in the total by taking the sum of the item, dividing it by the total, and multiplying by 100.
6. List the items being compared in decreasing order of the measure of comparison; e.g., the most frequent to the least frequent. The cumulative percent for an item is the sum of that item's percent of the total and that of all the other items that come before it in the ordering by rank.
7. List the items on the horizontal axis of a graph from highest to lowest. Label the left vertical axis with the numbers (frequency, time, or cost).
8. Label the right vertical axis with the cumulative percentages (the cumulative total should equal 100%).
9. Draw in the bars for each item.
10. Draw a line graph of the cumulative percentages. The first point on the line graph should line up with the top of the first bar.

# 2.4 Pareto Analysis

## *Example*

- Relative frequency of causes for errors on websites



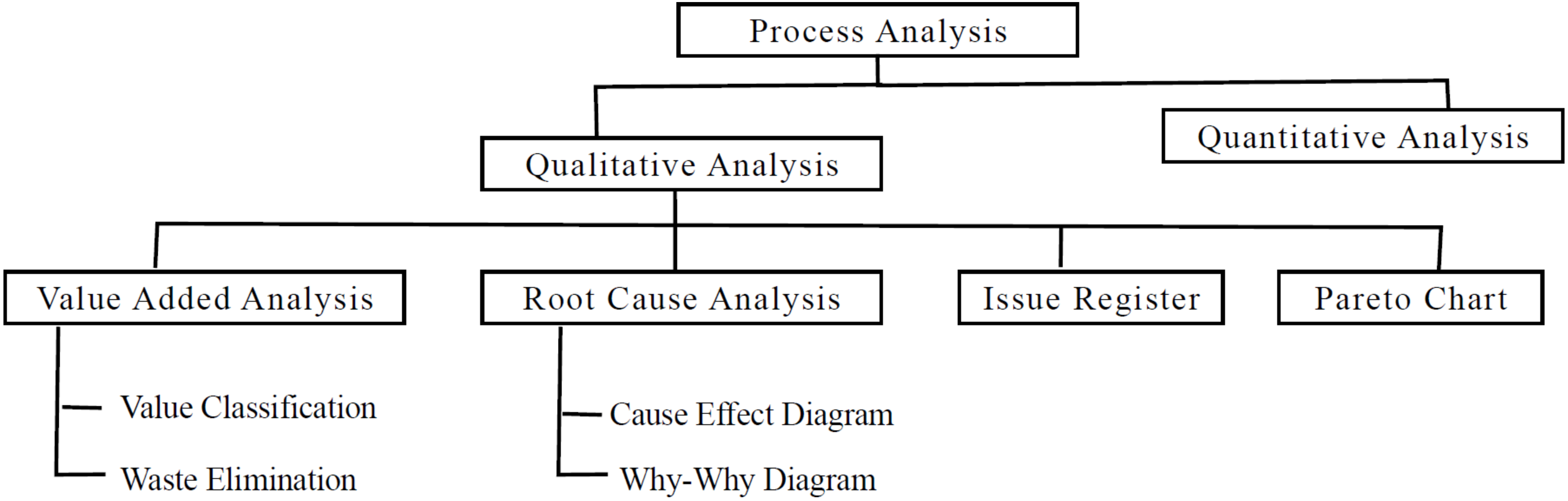

- It lets you see what 20% of cases are causing 80% of the problems and where you should focus efforts to achieve the most significant improvement.
- In this case, we can see that broken links, spelling errors and missing title tags should be the focus.

# 2.4 Pareto Analysis

## *Example*

| Quality Issues | Occurrences | Cumilative Precentage |
|---|---|---|
| Scratches | 40 | 40% |
| Broken Parts | 25 | 65% |
| Loose Fittings | 15 | 80% |
| Poor Finishing | 8 | 88% |
| Fabric Defects | 7 | 95% |

# Summary



Process Analysis
Qualitative Analysis
Quantitative Analysis
Value Added Analysis
Root Cause Analysis
Issue Register
Pareto Chart
Value Classification
Waste Elimination
Cause Effect Diagram
Why-Why Diagram

# TOPIC 03

Quantitative Process Analysis

If you had to choose between two services, you would typically choose the one that is:

If you had to choose between two services, you would typically choose the one that is:

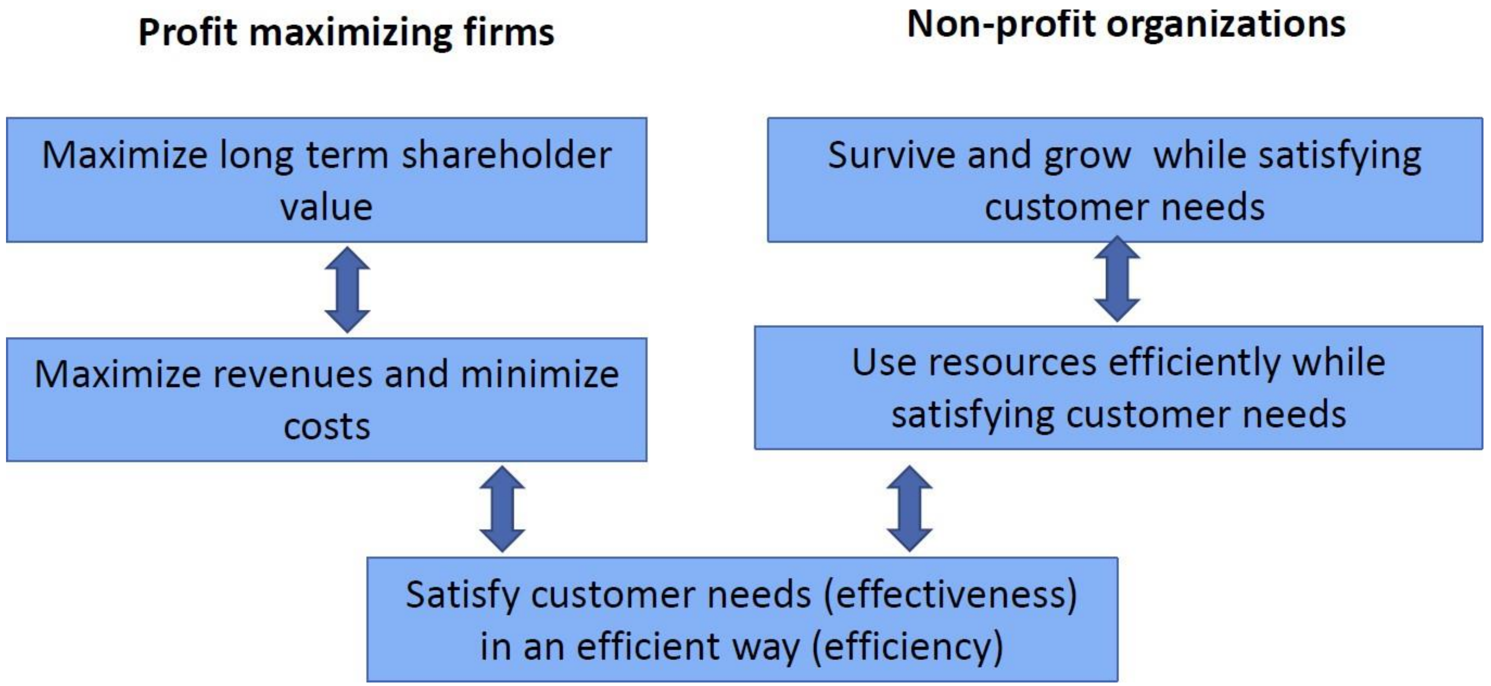
FASTER

BETTER

CHEAPER

# Process Performance Measures

- Link the identified processes to measurable objectives
- Quantify the benefits of improvement
- 2 types of performance measures;
  - Process Performance Dimensions
  - Balanced Scorecard



# Process Performance Measures

**Cost**
- Cost per execution
- Resource utilization
- Waste

**Time**
- Cycle time
- Waiting time / time spent in non-value-added tasks

**Quality**
- Error rates (negative outcomes, wrong info)
- Missed promise

# Techniques used for Quantitative Process Analysis

1. Flow Analysis
2. Queuing Analysis
3. Simulation

# 3.1 Flow Analysis

- Set of techniques that allows to estimate the overall performance of a process, given some knowledge about the performance of its activities
- Applications of flow analysis
  - Calculating Cycle time
  - Cycle time efficiency
  - Cycle time and work in progress
  - Calculating cost-per-process-instance
  - Calculating error rates at the process level
  - Estimating capacity requirements

# 3.1.1 Quantitative Flow Analysis - Cycle Time

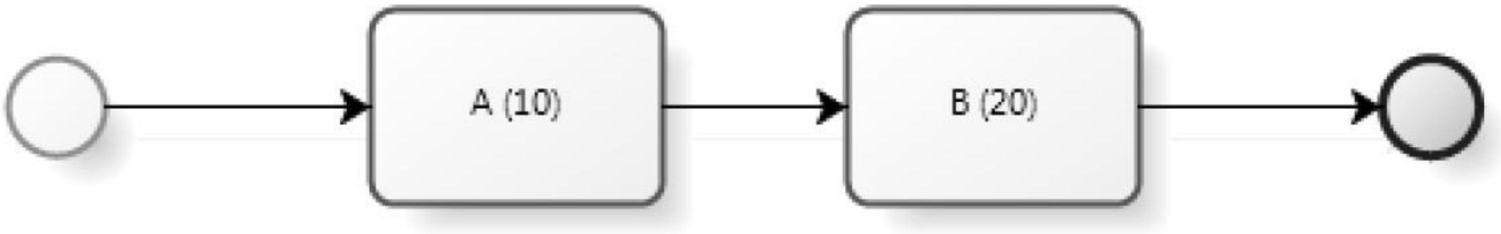
- **Cycle time**: Difference between a job's start and end time
- **Cycle time analysis**: the task of calculating the average cycle time for an entire process or process fragment
  - ✓ Assumes that the average activity times for all involved activities are available (activity time = waiting time + processing time)
- In the simplest case a process consists of a sequence of activities on a sequential path

  The average cycle time is the sum of the average activity times

  … but in general we must be able to account for

  - ✓ Alternative paths (XOR splits)
  - ✓ Parallel paths (AND splits)
  - ✓ Rework (cycles)

# 3.1.1 Quantitative Flow Analysis - Cycle Time
## Calculating Cycle time of a fully sequential process

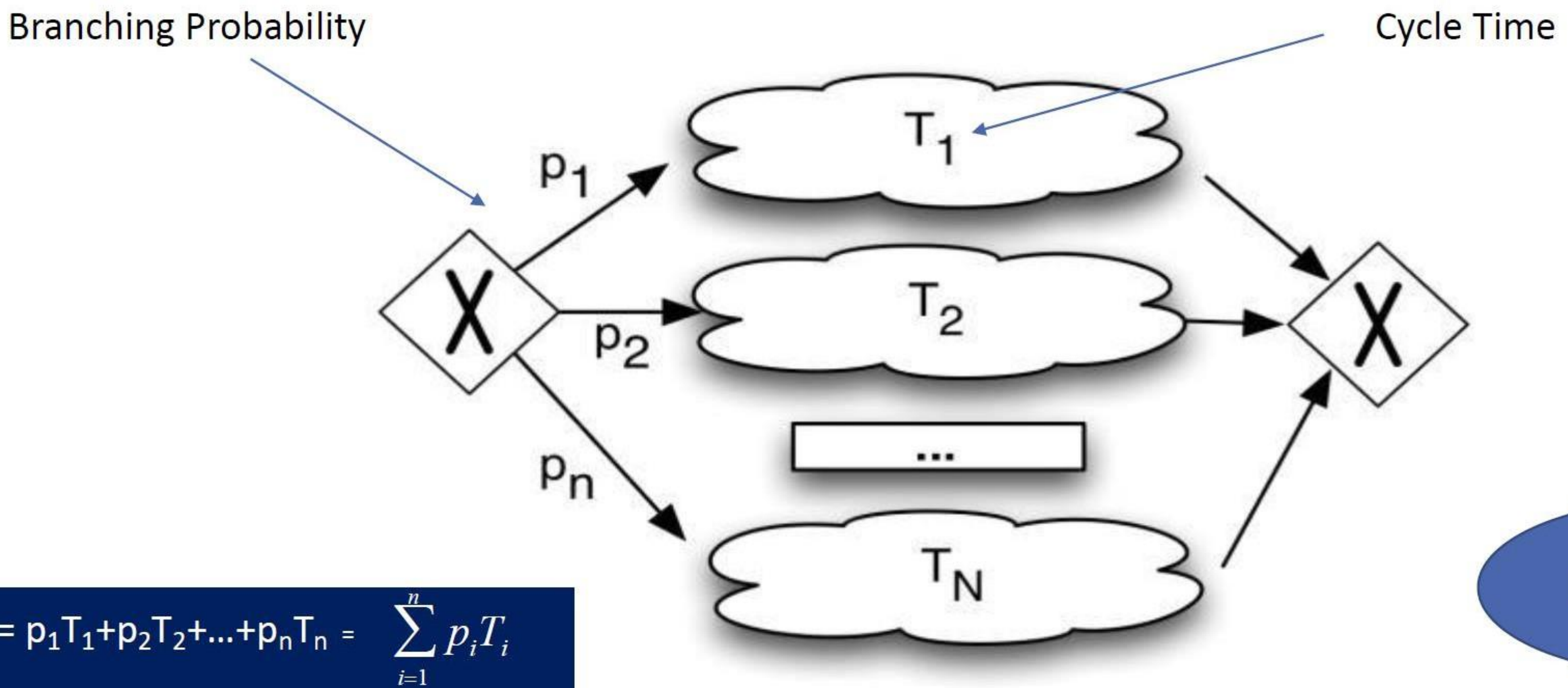



- Cycle time for each activity is indicated within brackets
- Since the activities are performed sequentially cycle time for this process;

$$10 + 20 = 30$$

# 3.1.1 Quantitative Flow Analysis - Cycle Time

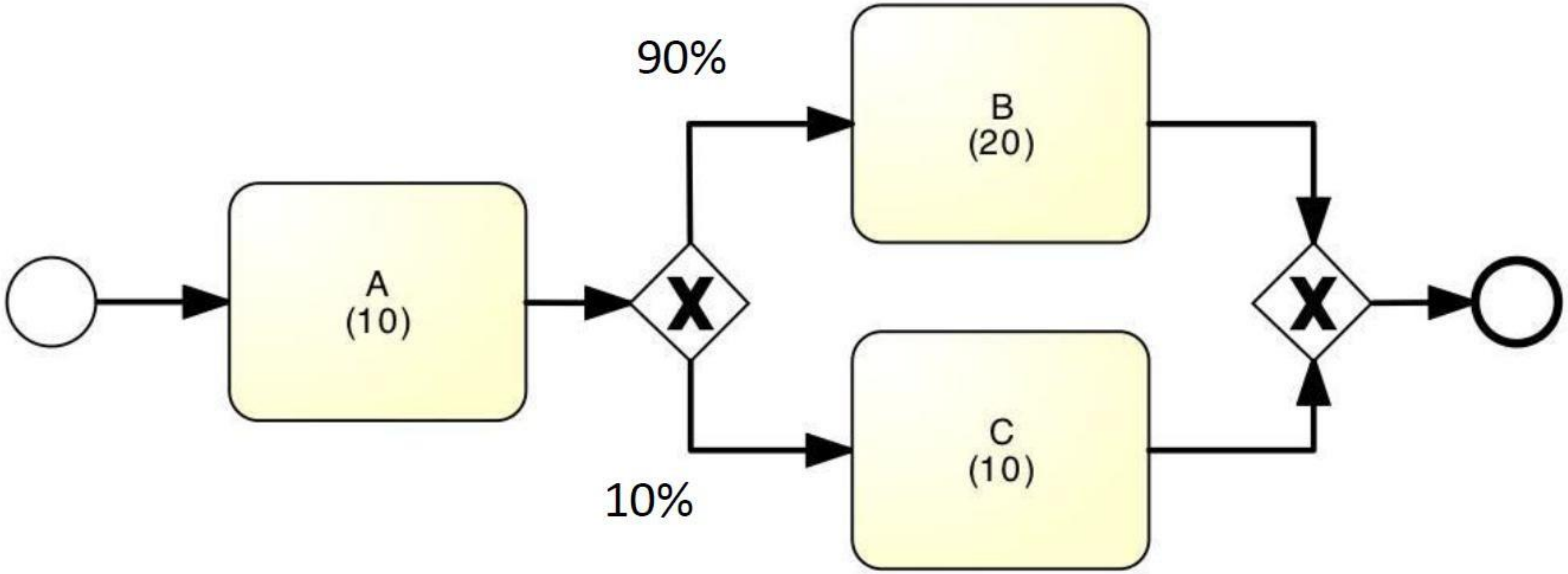
## Calculating Cycle time for alternative paths




$CT = p_1T_1 + p_2T_2 + \ldots + p_nT_n = \sum_{i=1}^{n} p_i T_i$

For XOR Split

# 3.1.1 Quantitative Flow Analysis - Cycle Time Example




Cycle time = cycle time of A + (90% * cycle time of B + 10% * cycle time of C)

CT = 10 + (0.9 * 20 + 0.1 * 10)

= 10 + (18 + 1)

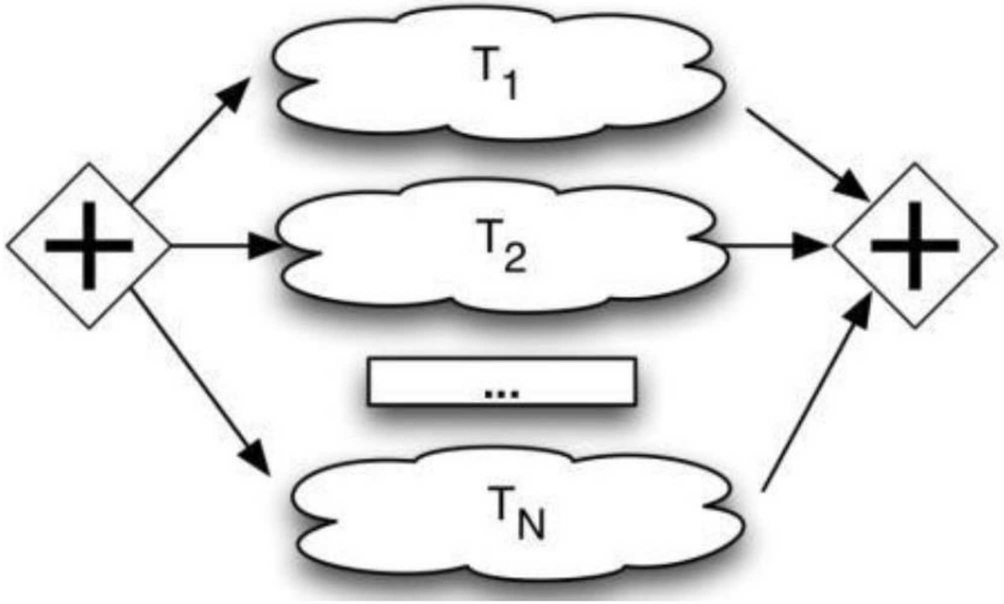
= 10 + 19

= 29

# 3.1.1 Quantitative Flow Analysis - Cycle Time

## Calculating Cycle time for parallel paths

If two activities related to the same job are done in parallel, the contribution to the cycle time for the job is the **maximum of the two activity times**.

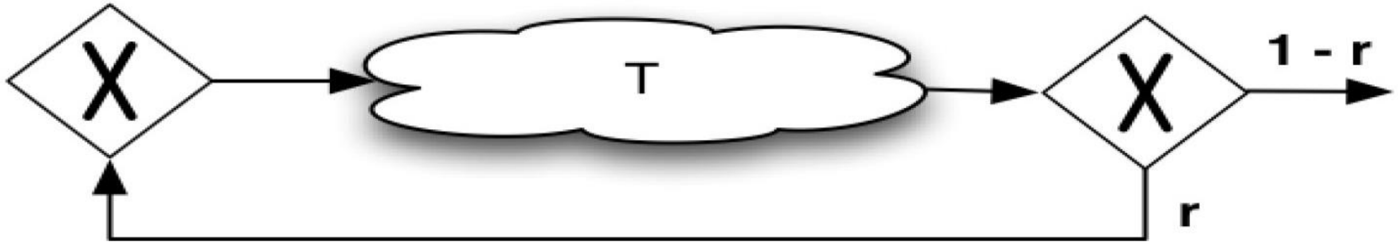



For AND Split

$CT_{parallel} = Max\{T_1, T_2, \ldots, T_M\}$

# 3.1.1 Quantitative Flow Analysis - Cycle Time Rework

- Sometimes a process has rework, which means an activity has to be repeated because of an error or failed inspection.
- In this case, the cycle time increases because the same work may need to be done more than once.
- The formula shown here helps estimate the average cycle time when rework is involved

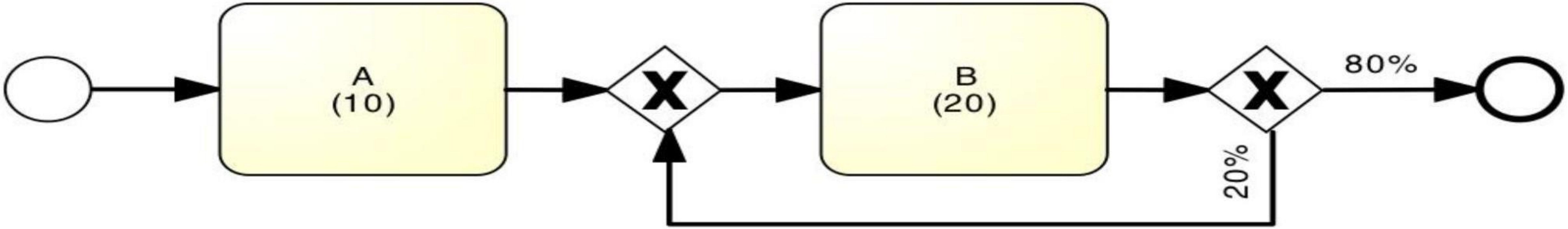



$$CT = T/(1-r)$$

**Cycle time = Cycle time of the Activity in the middle / Probability of the exit branch**

# 3.1.1 Quantitative Flow Analysis - Cycle Time Rework - Example

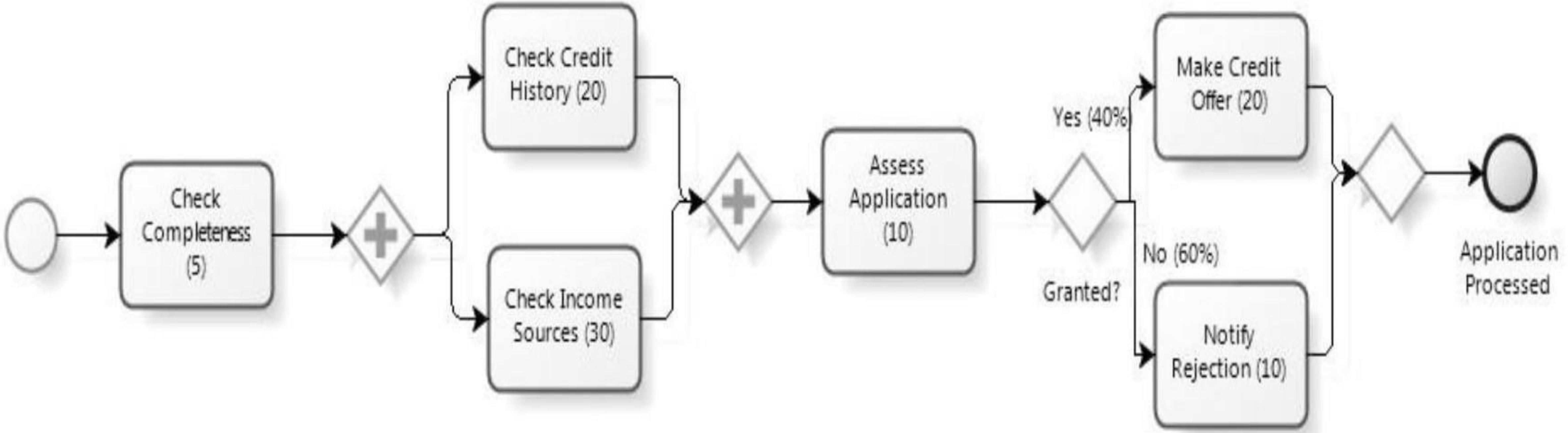
- What is the average cycle time?




Cycle time = 10 + (20/0.8) = 35

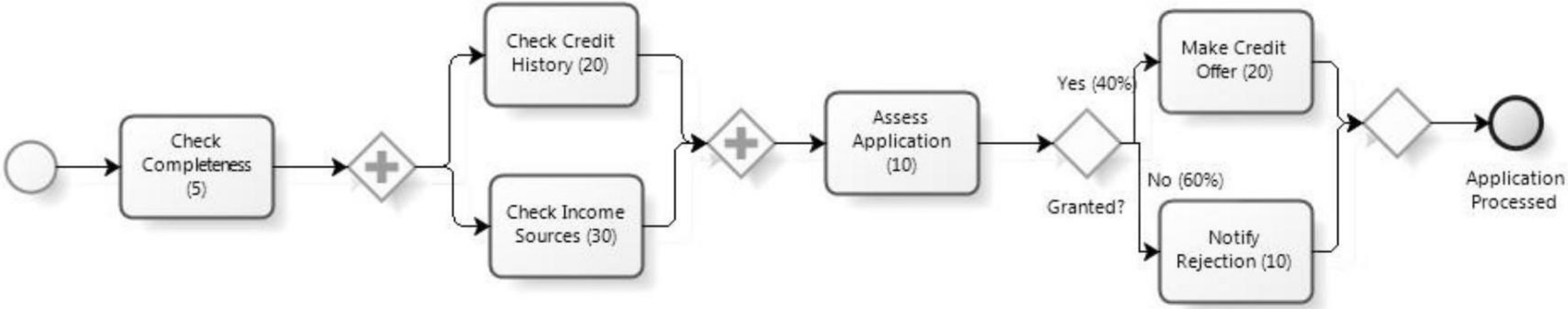
# 3.1.1 Quantitative Flow Analysis - Cycle Time Exercise

Calculate the CT for the given process



# 3.1.1 Quantitative Flow Analysis - Cycle Time Exercise

Calculate the CT for the given process




CT = 5 + 30 + 10 + (0.4*20 + 0.6 * 10)
= 5 + 30 + 10 + (8 + 6)
= 59

# 3.1.2 Quantitative Flow Analysis - Cycle Time Efficiency

- Measured as the percentage of the total cycle time spent on value adding activities.

$$\text{Cycle Time Efficiency} = \frac{\text{Theoretical Cycle Time}}{\text{CT}}$$

- Theoretical Cycle Time (TCT) is the cycle time if we only counted value-adding activities and excluded any waiting time or handover time
- Count only processing times
- CT = cycle time as defined before

**Measures the proportion of time spent on value-added activities in a process compared to the total time taken**

Example:

If a product takes 10 hours to manufacture, and only 2 hours are spent on actual production (value-added), the cycle time efficiency is (2 / 10) * 100% = 20%.

# 3.1.2 Quantitative Flow Analysis - Cycle Time And Work in Progress

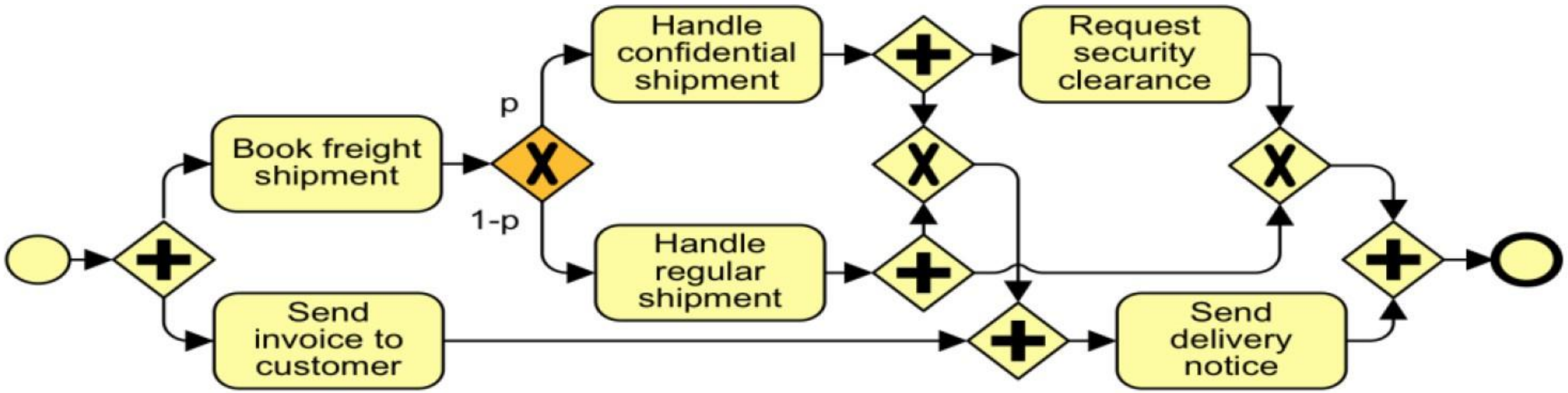
- WIP = (average) Work-In-Process
  - Number of cases that are running (started but not yet completed)
  - E.g. # of active and unfilled orders in an order-to-cash process

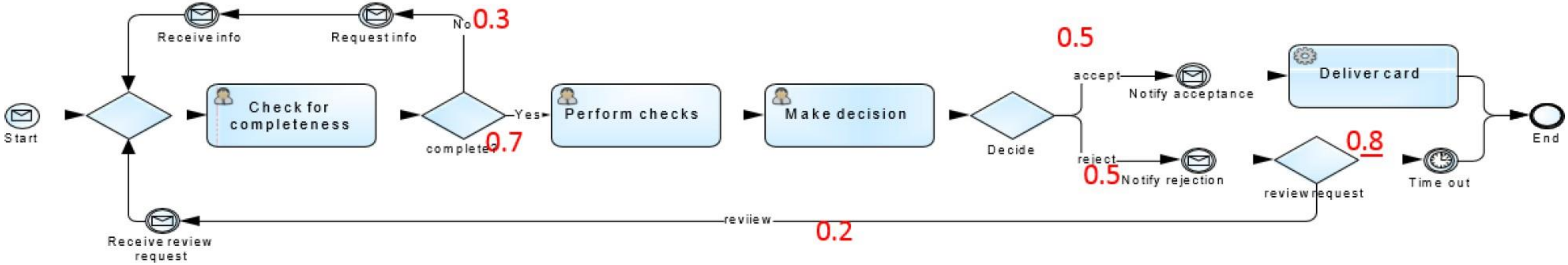
- Little's Formula: WIP = $\lambda \cdot$CT
  - $\lambda$ = arrival rate (number of new cases per time unit)
  - CT = cycle time

- WIP increases if the cycle time increases or if the arrival time increases.

# 3.1 Quantitative Flow Analysis

## Cycle Time Limitation 1 – Not all Models are Structured



# 3.1 Quantitative Flow Analysis

## Cycle Time Limitation 2 – Fixed load + Fixed resource

- Cycle time analysis does not consider waiting times due to resource contention
- Queuing analysis and simulation address these limitations and have a broader applicability

# Example - Cycle times for credit application process

## Activity Cycle time

Check completeness : 1 day (24 hours)
Check credit history : 1 day (24 hours)
Check income sources: 3 days (72 hours)
Assess application : 3 days (72 hours)
Make credit offer : 1 day (24 hours)
Notify rejection : 2 days (48 hours)

## Processing times

Check completeness: 2 hours
Check credit history: 30 minutes
Check income sources: 3 hours
Assess application: 2 hours
Make credit offer: 2 hours
Notify rejection: 30 minutes

- The table shows the cycle time and processing time for each activity.

# Example - Cycle times for credit application process

## Processing times

Check completeness: 2 hours
Check credit history: 30 minutes
Check income sources: 3 hours
Assess application: 2 hours
Make credit offer: 2 hours
Notify rejection: 30 minutes

The theoretical cycle time is the sum of the processing times for all the activities:

Theoretical Cycle Time = Sum of Activity Processing Times

**Theoretical Cycle Time** = 2 hours + 0.5 hours + 3 hours + 2 hours + 2 hours + 0.5 hours

**Theoretical Cycle Time** = 10 hours

# Example - Cycle times for credit application process

## Activity Cycle time

Check completeness : 1 day (24 hours)
Check credit history : 1 day (24 hours)
Check income sources: 3 days (72 hours)
Assess application : 3 days (72 hours)
Make credit offer : 1 day (24 hours)
Notify rejection : 2 days (48 hours)

The actual cycle time is the sum of the cycle times for all the activities:

Actual Cycle Time = Sum of Activity Cycle Times
**Actual Cycle Time** = 24 hours + 24 hours + 72 hours + 72 hours + 24 hours + 48 hours
**Actual Cycle Time** = 264 hours

# Example - Cycle times for credit application process

## Activity Cycle time

Check completeness : 1 day (24 hours)
Check credit history : 1 day (24 hours)
Check income sources: 3 days (72 hours)
Assess application : 3 days (72 hours)
Make credit offer : 1 day (24 hours)
Notify rejection : 2 days (48 hours)

## Processing times

Check completeness: 2 hours
Check credit history: 30 minutes
Check income sources: 3 hours
Assess application: 2 hours
Make credit offer: 2 hours
Notify rejection: 30 minutes

Cycle Time Efficiency = Theoretical Cycle Time / Actual Cycle Time

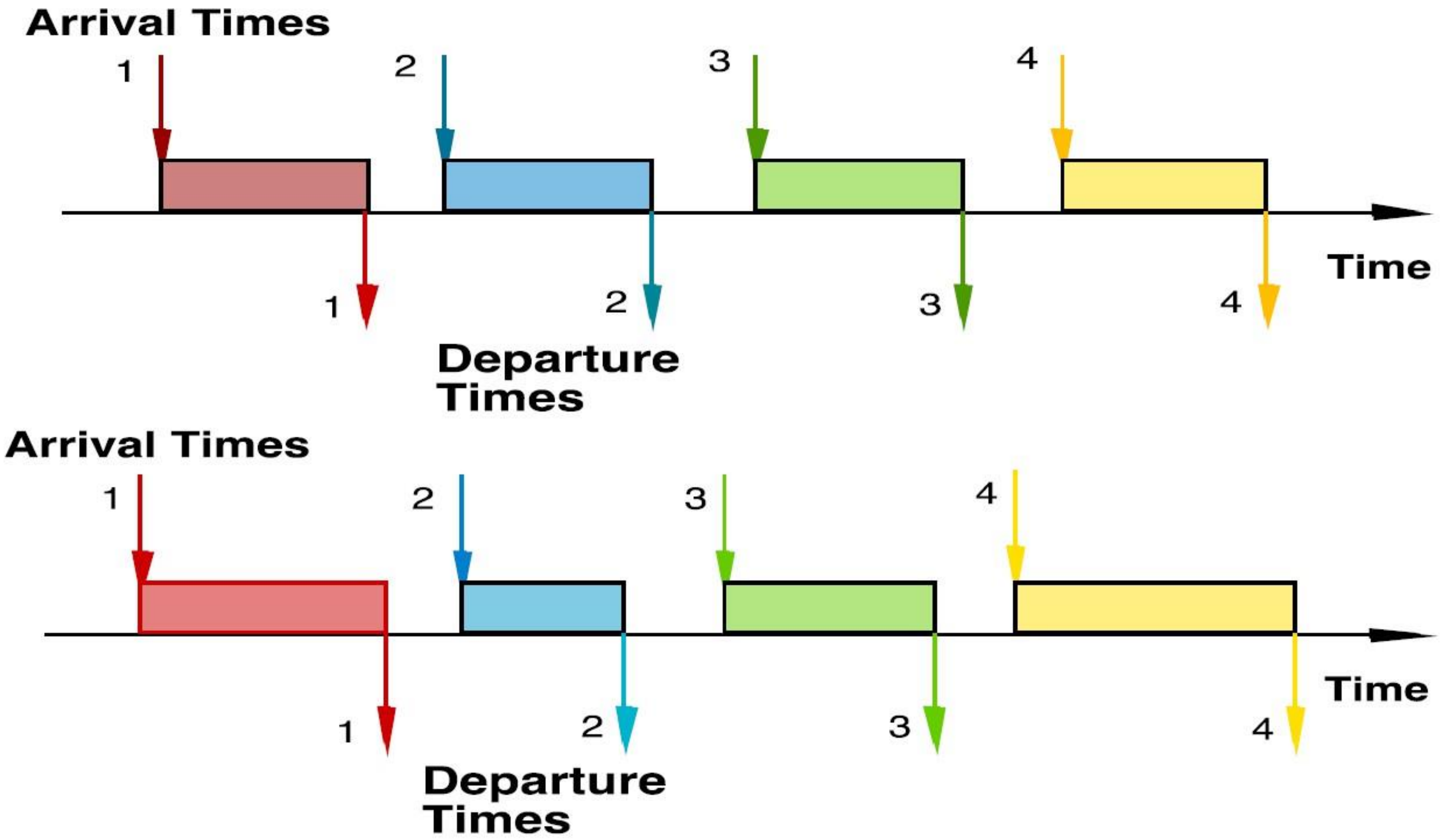
**Cycle Time Efficiency** = 10 hours / 264 hours ≈ 0.0379 or 3.79%

In this case, the cycle time efficiency is approximately **3.79%,** which means that the **actual cycle time** is significantly longer than the **theoretical cycle time.** This indicates that there are delays and inefficiencies in the process, leading to a **low cycle time efficiency**.

# 3.2 Queuing Theory

- A collection of mathematical techniques to analyze systems that have resource contention/conflict
- Resource contention leads to queues (as in supermarkets)
- Capacity problems are very common in industry and one of the main drivers of process redesign
- Need to balance the cost of increased capacity against the gains of increased productivity and service
- Queuing and waiting time analysis is particularly important in service systems
    - Large costs of waiting and of lost sales due to waiting Queuing Theory

# 3.2.1 Queuing Theory - Delay is Caused by Job Interference




Deterministic traffic

Variable but spaced apart traffic

- If arrivals are regular or sufficiently spaced apart, no queuing delay occurs

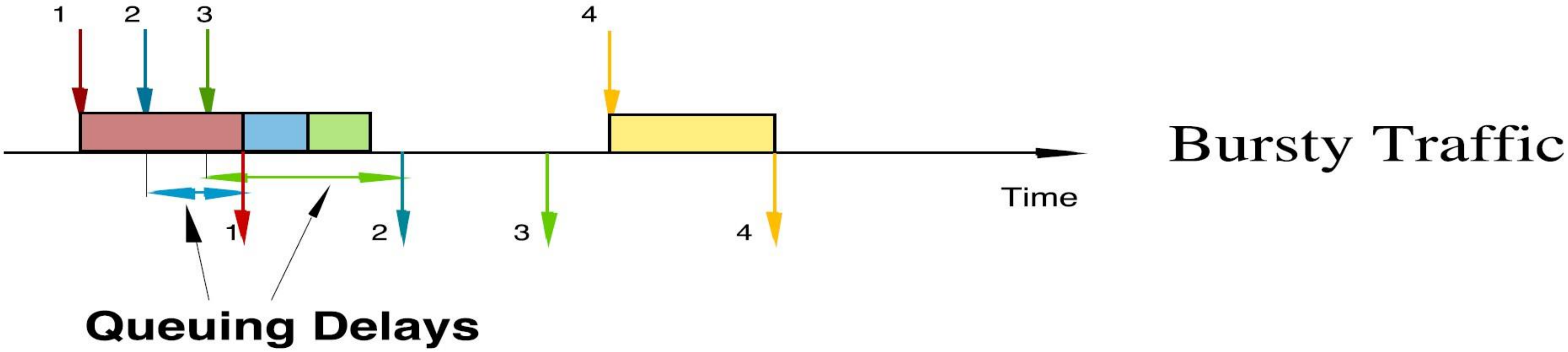
# Delay is Caused by Job Interference

"When multiple jobs or tasks are competing for the **attention** of the same **resources** (such as servers or processors), **interference** can occur. This interference leads to delays as jobs wait for their turn to be processed. When resources are **shared**, conflicts arise, and jobs may have to wait longer due to the **interference caused** by other jobs."

Imagine there is a **single cashier handling customers**. If a customer at the front of the line has a **complicated return process** (such as needing to verify a receipt, fill out forms, etc.), this interferes with the service of other customers waiting to check out.

# 3.2.2 Queuing Theory - Burstiness Causes Interference

* Queuing results from variability in service times and/or interarrival intervals

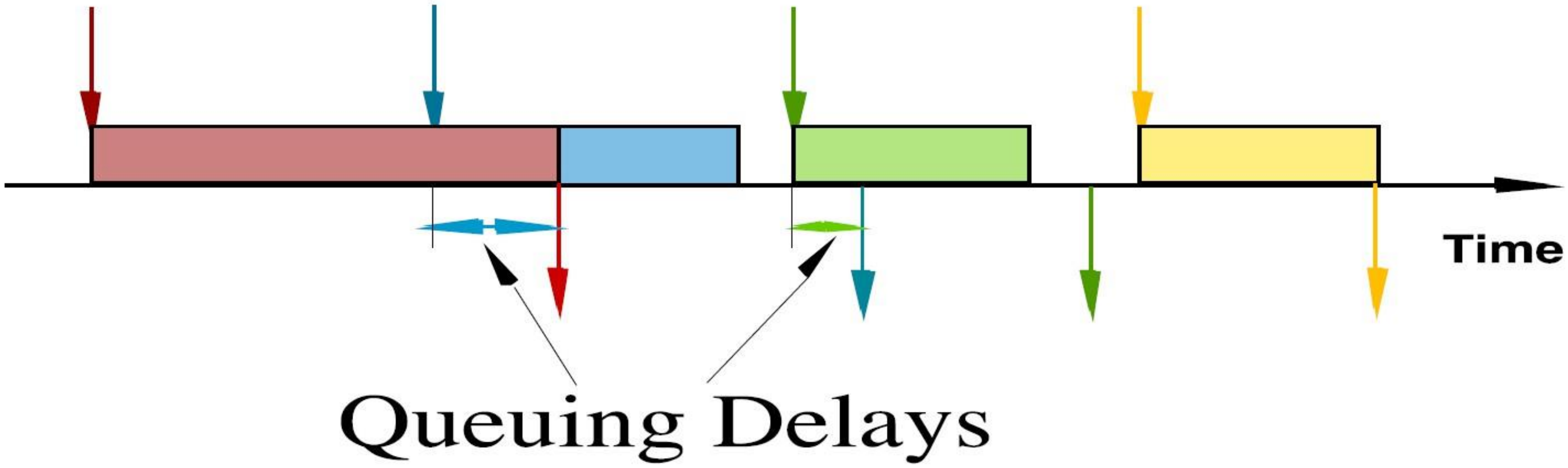


# Burstiness Causes Interference

"Burstiness refers to the irregular pattern of arrivals or service times in a queueing system. When arrivals or service demands are **bursty**, it can lead to variations in queue lengths and waiting times, causing interference among jobs.."

At the supermarket, burstiness might occur right after a **local event** (e.g., a sports game ends, and many people come to the store at once). This sudden surge in customers creates interference at the checkout counters, overwhelming the cashiers and causing long waits, even if the service rate is typically sufficient during normal times.

# 3.2.3 Queuing Theory - Job Size Variation Causes Interference




- Deterministic arrivals, variable job sizes

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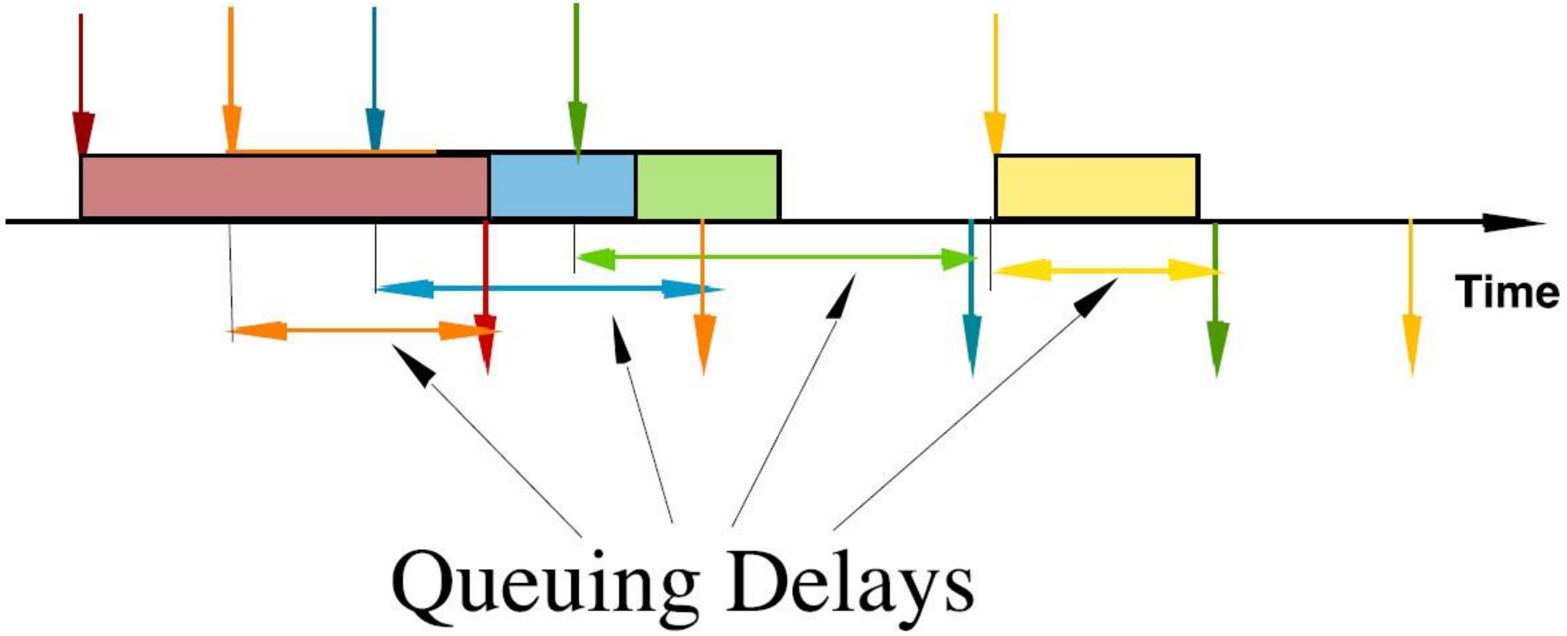
# Job Size Variation Causes Interference

"When jobs have **varying sizes or service time requirements**, it can lead to interference as shorter jobs might get **delayed** due to longer jobs taking up resources for **extended periods**."

Consider the variation in checkout times based on the **number of items** customers have. One customer might have only a few items, while another has a full cart. If a customer with a full cart is being processed, it will take significantly longer than a customer with just a few items, causing delays for those in line behind the customer with the full cart.

# 3.2.4 Queuing Theory - High Utilization Exacerbates Interference




- The queuing probability increases as the load increases
- Utilization close to 100% is unsustainable → too long queuing times

# High Utilization Exacerbates Interference

"When the system's resources are **heavily utilized**, interference among jobs becomes more pronounced. High utilization means that resources are **consistently busy**, making it challenging for jobs to get timely service and leading to longer waiting times."

If the supermarket is operating at **near full capacity**—let's say each cashier is almost always busy—there's little room to handle unexpected situations (like a price check, customer disputes, or burstiness in customer arrivals). Even a small delay, like needing to call a **manager for assistance**, can cascade into significant delays for all customers in line because there are no idle cashiers to step in and help.

# Shared Resources Introduce Interference

"When resources, such as **servers** or **channels**, are shared among different jobs or tasks, interference can occur when one **job's usage** affects another's performance. Sharing resources often leads to **contention** and **queuing**, causing delays for jobs waiting for resource access."

Imagine the supermarket has only **one card payment machine** that's shared between **two checkout counters**. If one cashier is processing a card payment, the other cashier has to wait until the machine is available.

# 3.2.5 Queuing Theory

## *Example*

- Patients arrive by ambulance or by their own accord
- One doctor is always on duty
- More patients seeks help -longer waiting times

Question: Should another position be instated?

# 3.2.5 Queuing Theory - Drawbacks of queuing theory

- Limited Representation of Real-World Factors: Queuing models may not fully capture real-world complexities such as customer behavior, system interruptions, priority levels, and complex service patterns.

- Generally not applicable when system includes parallel activities.

- Dynamic Nature of Systems: Many systems are dynamic, with changing conditions over time. Queuing models might not effectively capture how system performance evolves in response to changing conditions.

# 3.3 Process Simulation

- Process Simulator generates a large number of hypothetical instances of a process, execute step-by-step and record each step of execution.
- Run a large number of process instances, gather data (cost, duration, resource usage) and calculate statistics from the output.
- The output of a simulator is includes logs of the simulation as well as statistics related to the cycle times, waiting times and resource utilization.
- Process simulation is more versatile (also more popular).

# 3.3.1 Process Simulation - Steps in evaluating a process with simulation

- Model the process (e.g. BPMN)
- Enhance the process model with simulation info -----simulation model Based on assumptions or better based on data (logs)
- Run the simulation
- Analyze the simulation outputs
  - Process duration and cost stats and histograms
  - Waiting times (per activity)
  - Resource utilization (per resource)
- Repeat for alternative scenarios

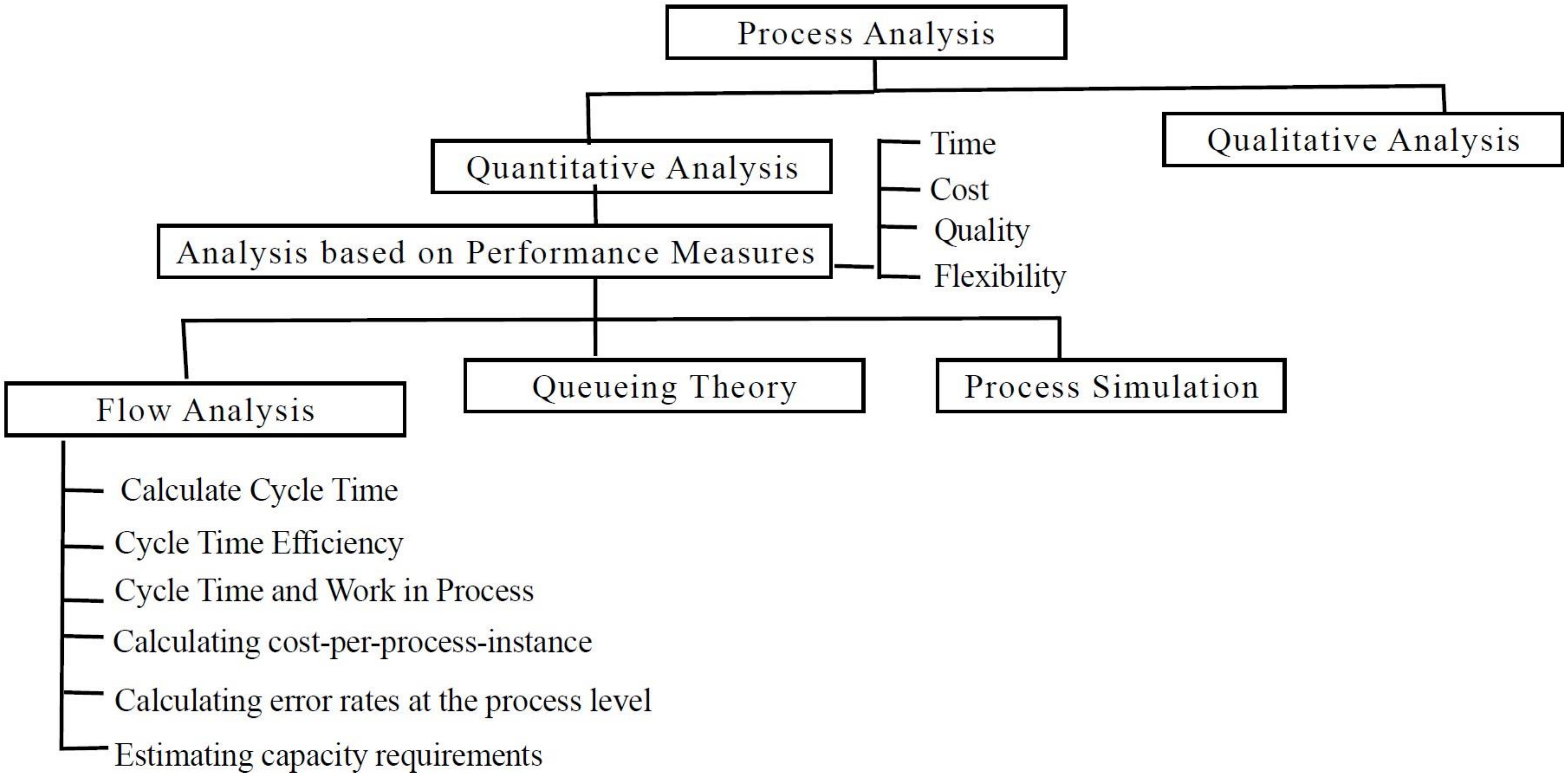
# 3.3.2 Process Simulation - Elements of a simulation model

- The process model including:
  - Events, activities, control-flow relations (flows, gateways)
- Resource classes (i.e. lanes)
  - Resource assignment ----Mapping from activities to resource classes
- Processing times
  - Per activity or per activity-resource pair
- Costs
  - Per activity and/or per activity-resource pair
- Arrival rate of process instances
- Conditional branching probabilities (XOR gateways)

# 3.3.3 Process Simulation - Tools for Process Simulation

- Visio
- ARIS
- Online tool: BIMP: http://bimp.cs.ut.ee/ (Accepts standard BPMN 2.0 asinput)

# Summary



Process Analysis
Quantitative Analysis
Qualitative Analysis
Analysis based on Performance Measures
Time
Cost
Quality
Flexibility
Flow Analysis
Queueing Theory
Process Simulation
Calculate Cycle Time
Cycle Time Efficiency
Cycle Time and Work in Process
Calculating cost-per-process-instance
Calculating error rates at the process level
Estimating capacity requirements


# Recap…

# Thank You!